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SoftFuse-KGC: Soft Support and Confidence-Aware Fuzzy Retrieval for Biomedical Knowledge Graph Completion

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Files for peer review

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Name	Type of File	Size	Page
Supplementary Materal.pdf	Supplementary Material for Review	912.1 KB	Page 3
Graphical Abstract Image.png	Graphical Abstract Image	170.9 KB	Page 20
Main Manuscript.pdf	Main Document - LaTeX PDF	584.0 KB	Page 21

Supplementary Materal

The Web Application

We provide a public web application for interactive inspection of the SoftFuse-KGC pipeline at: https://inosuke710-softfuse-kgc-demo.hf.space/. The web application is a demonstration deployed on Hugging Face Spaces and uses precomputed artifacts. It allows users to inspect validation and test examples across PrimeKG, PharmKG, Hetionet, DRKG, and repoDB. After opening the page, users select a dataset, split, disease/query case, and top-k value. The three main tabs then expose the complete demo flow: an overall prediction summary, candidate reranking by soft support, and evidence-subgraph selection by fuzzy retrieval.

Figures S1–S3 document the three tabs and identify the controls, result panels, candidate lists, graph elements, and visual encodings used in the interface.

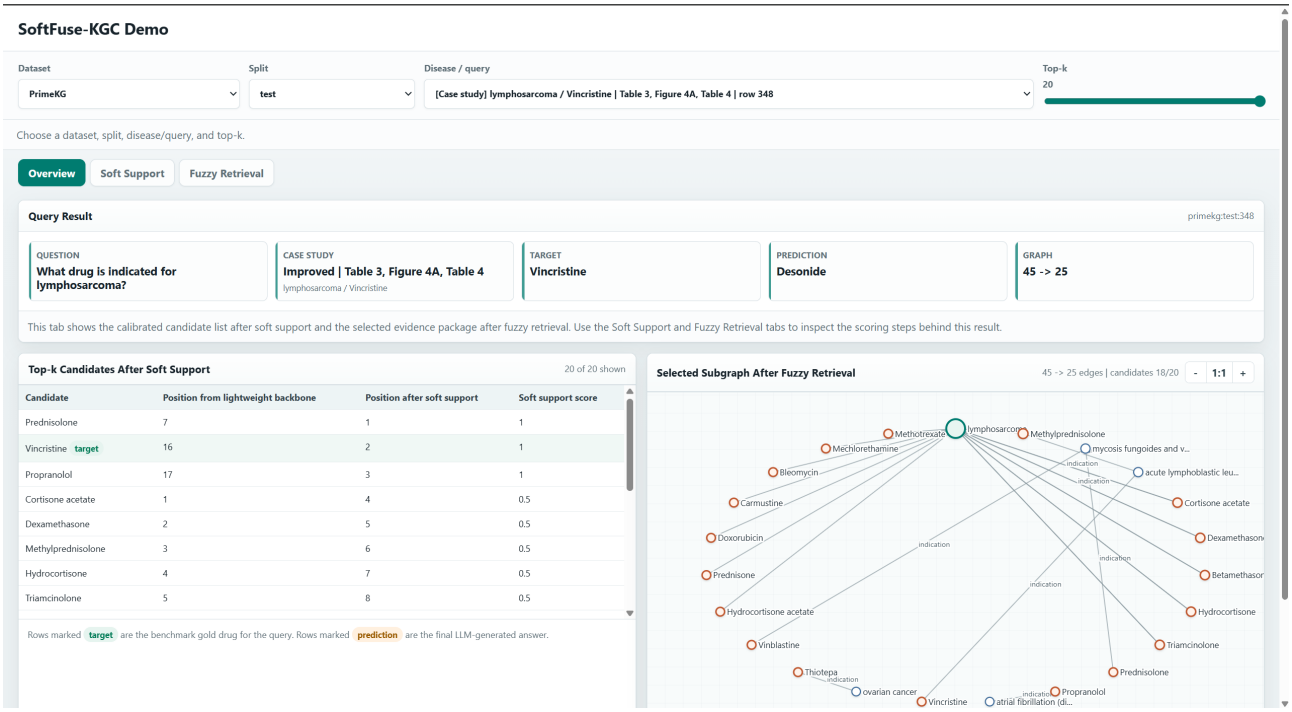


Fig. S1 Overview tab. Controls select the dataset, evaluation split, disease or query case, and displayed top-k. The Query Result panel reports the question, case-study label when available, gold target, prediction, and graph-compression size. The candidate table gives backbone rank, calibrated rank, and soft-support score, with text markers for the target and prediction. The graph panel shows the selected fuzzy evidence subgraph and provides zoom, reset, and drag controls

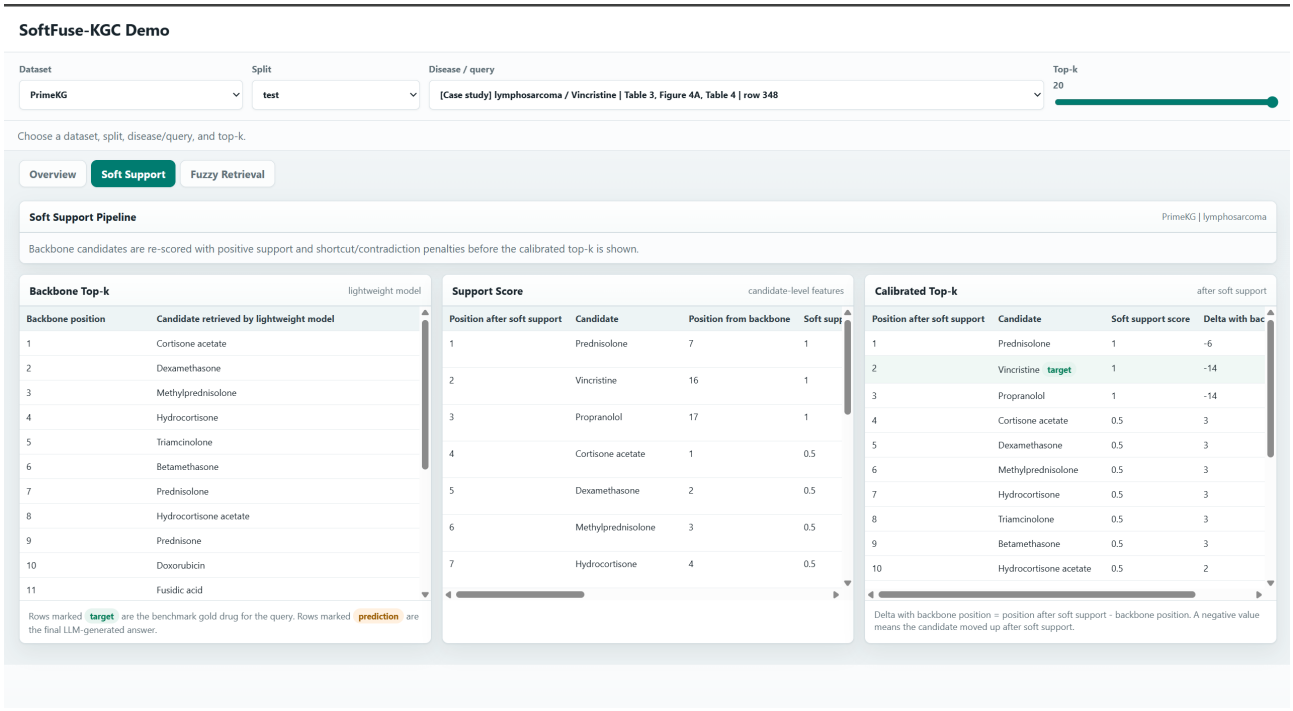


Fig. S2 Soft Support tab. The left panel reports the original lightweight-backbone top-k order. The center panel shows the final support score and available feature terms, including positive evidence, shared-gene or bridge evidence, and shortcut or contradiction penalties. The right panel gives the calibrated top-k. "Delta with backbone position" is calibrated rank minus backbone rank; negative values indicate upward movement

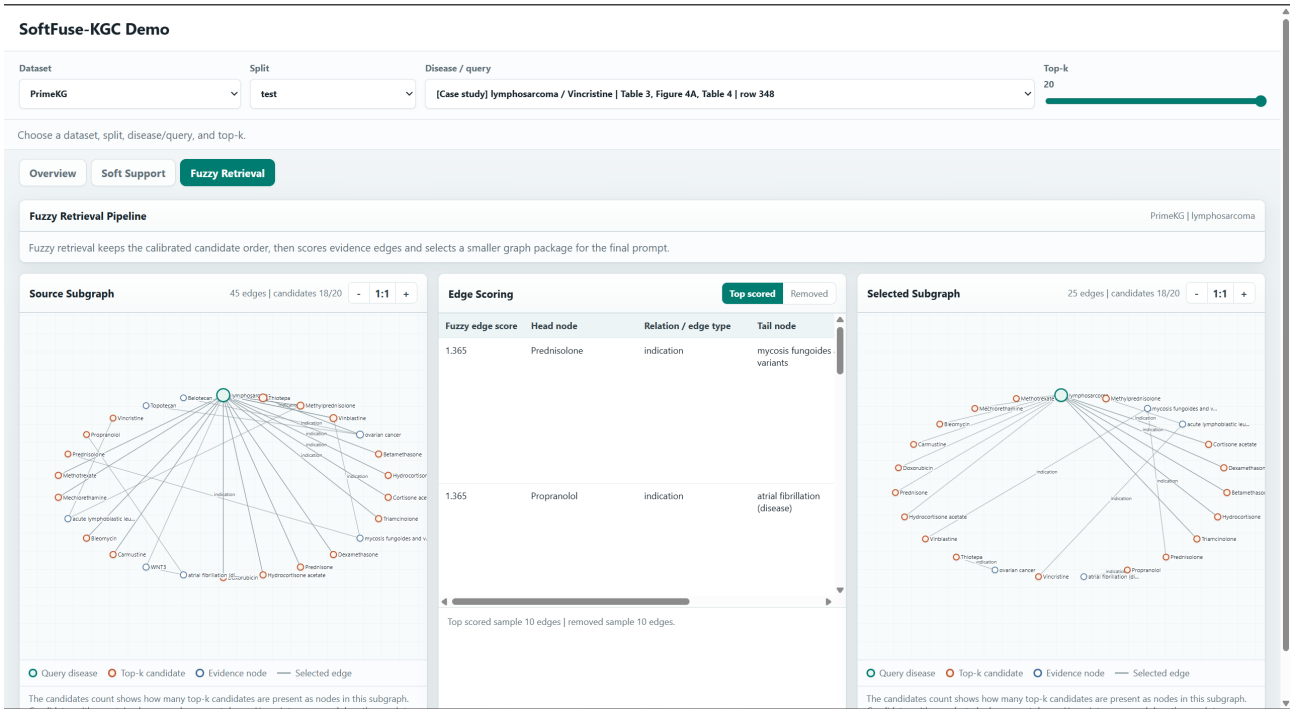


Fig. S3 Fuzzy Retrieval tab. The left panel reports the source subgraph, source-edge count, and candidate coverage. The center panel lists fuzzy edge scores and their component terms and can show either retained or removed evidence. The right panel shows the selected compact subgraph. Labels and the legend identify the query disease, top-k candidates, evidence nodes, and selected edges; color is a redundant visual cue. Zoom, reset, and drag controls support inspection

S1 Soft-Support Feature and Weight Assignment

Equation numbers in this Online Resource refer to the corresponding equations in the main article.

$$
\begin{aligned}
\text{Eq. (2):}\quad E_i(q) &= \omega_{\text{touch}} T_i(q) + \omega_{\text{path}} P_i(q) + \omega_{\text{schema}} A_i(q) + \omega_{\text{density}} \Delta_i(q),\\
\text{Eq. (3):}\quad P_i(q) &= \nu_s M_i^s(q) + \nu_b M_i^b(q)\\
&\quad + \nu_c T_i^c(q) + \nu_q T_i^q(q),\\
M_i^s(q) &= \min\left(1, \frac{|G(c_i) \cap G(q)|}{\eta_s}\right),\\
M_i^b(q) &= \min\left(1, \frac{B_i(q)}{\eta_b}\right),\\
\text{Eq. (5):}\quad \Delta_i(q) &= \min\left(1, \frac{\log(1 + \deg_{S_q^{(0)}}(c_i) + \deg_{S_q^{(0)}}(t_q))}{\eta_\Delta}\right),\\
\text{Eq. (6):}\quad D_i(q) &= \max\left(\mathbb{I}[c_i \leftrightarrow t_q \text{ in } S_q^{(0)}], \min\left(1, \frac{\deg_{\text{direct}}(c_i)}{\eta_D}\right)\right),\\
\text{Eq. (7):}\quad X_i(q) &= \mathbb{I}[c_i \leftrightarrow t_q \text{ through contradiction-like evidence}],\\
\text{Eq. (8):}\quad \widetilde{S}_{\theta_d}(c_i, q) &= \alpha_{E,d} E_i(q) - \alpha_{D,d} D_i(q) - \alpha_{X,d} X_i(q),\\
S_{\theta_d}(c_i, q) &= \mu_d R_i(q) + (1 - \mu_d) N_q(\widetilde{S}_{\theta_d}(c_i, q)).
\end{aligned}
$$

When $\mu_d > 0$, the within-query normalizer in Eq. (8) is

$$
N_q(z_i) = \begin{cases} \dfrac{z_i - \min_j z_j}{\max_j z_j - \min_j z_j}, & \max_j z_j > \min_j z_j,\\ 0.5, & \max_j z_j = \min_j z_j. \end{cases}
$$

S1.1 Feature Selection Method

The soft-support feature family was obtained through an audit-and-calibration procedure. First, for each validation query and each top-20 candidate, a candidate-level feature table was built from the retrieved source subgraph and the training evidence graph. The initial table included type/schema flags, ontology-support flags, candidate evidence contact, query evidence contact, direct candidate-query contact, contradiction or conflict flags, and local count features. For mechanistic datasets, the same procedure was extended to compound-gene, disease-gene, and gene-gene neighborhoods, producing shared-gene, bridge, candidate-touch, and query-touch counts.

The candidate attributes were then summarized by frequency, value range, and gold-versus-non-gold separation. Attributes that were constant or nearly constant on the candidate set were not used as active ranking signals. Attributes that were too sparse but had a clear negative or diagnostic interpretation were retained only as small penalty channels. Attributes that appeared often enough and had an interpretable biomedical role were grouped into the general formula: $T_i(q)$ for evidence contact, $P_i(q)$ for mechanism-oriented support, $A_i(q)$ for schema compatibility when informative, $\Delta_i(q)$ for bounded local density, $D_i(q)$ for direct shortcut or hub-like evidence, and $X_i(q)$ for contradiction-like evidence.

Candidate-stage formulas were then probed as simple score variants before freezing the final coefficient family. The validation rule was to preserve Gold@20, improve or maintain MRR@20, and reduce shortcut exposure. This procedure is why the final soft-support formula separates positive evidence from direct-link penalties instead of using a hard ontology filter or a raw count score. On PrimeKG, for example, the selected contact-based b050 row improved MRR@20 to 0.1356, had no worsened queries relative to the backbone, and reduced the top-five direct-link rate from 0.2408 to 0.0072. The count-like capped-evidence contrast dropped to $\text{MRR@20} = 0.0925$ and restored a shortcut-heavy top-five direct-link rate of 0.2452.

The candidate-density feature in Eq. (5) is retained in the general formulation because it was part of this audited feature family, but it is inactive in the reported candidate reranking rows. In the validation audit, local density

behaved as a hub-sensitive count signal at the candidate stage rather than as a reliable therapeutic-support signal. It is therefore assigned zero candidate-stage weight and retained later only as a small edge-level tie breaker in Eq. (13).

S1.2 Weight Assignment Method

Weights and normalizers were fixed using validation data only. Candidate-stage settings were retained when they preserved Gold@20 and improved or maintained MRR@20 while reducing shortcut exposure. For Eq. (3), η_s and η_b are saturation thresholds for shared-gene and bridge counts, and $\nu_s, \nu_b, \nu_c, \nu_q$ distribute positive support across shared-gene, bridge, candidate-touch, and query-touch channels. For Hetionet and DRKG, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$.

For Eq. (6), η_D is the 95th percentile of nonzero compound-to-disease direct degrees when direct-degree statistics are used. This high-percentile normalizer penalizes generic direct-disease hubs without letting a single extreme hub define the scale. Otherwise, the direct edge flag alone is used. For Eq. (8), N_q is the identity map when $\mu = 0$; when $\mu > 0$, the min–max rule above is applied over the 20 candidates for the same query.

S1.3 Frozen Assigned Values

Table S1 reports the frozen feature instantiations, weights, and normalizers.

Table S1 Dataset-specific soft-support coefficients and feature instantiations. These values instantiate Eq. (2), Eq. (3), Eq. (5), Eq. (6), Eq. (7), and Eq. (8).

Dataset	Positive support instantiation	Penalty and rank-prior coefficients	Normalizers and inactive terms
PrimeKG	In Eq. (2), $E_i(q)$ is contact-based: $E_i(q) = 1$ when the candidate has positive evidence contact in the source subgraph and 0 otherwise.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (6), $D_i(q)$ is the direct candidate-query flag; in Eq. (7), $X_i(q)$ uses contradiction-like evidence when available.	The path term in Eq. (3), density term in Eq. (5), and direct-degree normalizer in Eq. (6) are inactive in the final candidate score.
PharmKG	In Eq. (2), $E_i(q)$ is the evidence-positive support flag for the therapeutic-association proxy relation.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (7), $X_i(q)$ is fixed to zero because no stable contradiction-like relation is used.	The path term in Eq. (3), density term in Eq. (5), and direct-degree normalizer in Eq. (6) are inactive in the final candidate score.
Hetionet	In Eq. (3), mechanistic support combines shared genes, gene-gene bridge evidence, candidate touch, and query touch. Candidate-gene cap $= 120$, disease-gene cap $= 300$, bridge cap $= 30$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.50$, $\alpha_X = 0.10$, and $\mu = 0$. In Eq. (6), direct shortcut evidence includes direct drug-disease evidence and generic direct-disease degree.	In Eq. (3), $\eta_s = 2$, $\eta_b = 5$, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$. In Eq. (6), $\eta_D = 5.0$. The candidate-density term in Eq. (5) is inactive.
DRKG	In Eq. (3), mechanistic support combines shared genes, gene-gene bridge evidence, candidate touch, and query touch. Candidate-gene cap $= 120$, disease-gene cap $= 240$, bridge cap $= 40$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0.10$, $\alpha_X = 0.10$, and $\mu = 0.80$. The rank prior is retained because support features are sparse and hub-sensitive.	In Eq. (3), $\eta_s = 2$, $\eta_b = 6$, $\nu_s = 0.70$, $\nu_b = 0.20$, $\nu_c = 0.05$, and $\nu_q = 0.05$. In Eq. (6), $\eta_D = 52.95$. The candidate-density term in Eq. (5) is inactive.
repoDB	In Eq. (2), diagnostic support combines approved direct evidence, compound-gene evidence, and gene-bridge evidence: 0.55 approved direct, $0.30\,\min(1, \text{compound gene}/8)$, and $0.15\,\min(1, \text{gene bridge}/12)$.	In Eq. (8), $\alpha_E = 1.0$, $\alpha_D = 0$, $\alpha_X = 0.20$, and $\mu = 0.95$. The high rank-prior mixture reflects sparse query-specific mechanistic support in the local disease namespace.	The direct-degree feature in Eq. (6) is computed but inactive because $\alpha_D = 0$. The candidate-density term in Eq. (5) is inactive. In Eq. (7), the failed/diagnostic-like penalty is retained as $X_i(q)$.

S2 Fuzzy Edge-Scoring Feature and Weight Assignment

$$\text{Eq. (10):} \quad B(e,q) = \max_{c_i \in C_q^{\text{in}}: e \sim c_i} w_b(r_i),$$

$$w_b(r_i) = \begin{cases} 1.00, & 1 \le r_i \le 5, \\ 0.60, & 6 \le r_i \le 10, \\ 0.25, & 11 \le r_i \le 20, \end{cases}$$

$$
\begin{aligned}
\text{Eq. (11):}\quad S_{\text{edge}}(e,q) &= \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\
&\quad + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\
&\quad - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \\
\text{Eq. (12):}\quad T_e^c(q) &= \mathbb{I}[e \text{ touches at least one candidate}], \\
T_e^q(q) &= \mathbb{I}[e \text{ touches the query entity}], \\
H_e(q) &= \mathbb{I}[e \text{ touches a top-band candidate}], \\
\text{Eq. (13):}\quad \rho_e(q) &= \frac{\log(1+\deg_{S_q^{(0)}}(u)) + \log(1+\deg_{S_q^{(0)}}(v))}{\eta_\rho}.
\end{aligned}
$$

S2.1 Feature Selection Method

The fuzzy-retrieval feature family was selected after the candidate order had been fixed. The first step was to convert each retrieved source subgraph into a triple-level feature table. For each evidence triple, the audit recorded whether it touched a candidate, which candidate rank band it touched, whether it touched a top-band candidate, whether it touched the query entity, whether it formed a direct candidate-query shortcut, whether it inherited a contradiction-like flag, its relation family when relation labels were stable, and a bounded local-density hint.

These triple attributes were summarized by occurrence frequency and by their effect on coverage-preserving graph compression. Features were retained when they controlled one of the retrieval decisions needed by the fuzzy package: preserving candidate and query anchoring, transferring confidence from the fixed candidate ranking, prioritizing interpretable biomedical relation families, demoting direct or negative-like shortcut evidence, or breaking ties among otherwise comparable triples. This audit produced the candidate-band score $B(e,q)$, incidence indicators $T_e^c(q)$ and $T_e^q(q)$, the top-candidate indicator $H_e(q)$, relation-family priority $F_r(e)$, shortcut and contradiction penalties $D_e(q)$, $X_e(q)$, and the density term $\rho_e(q)$.

The resulting features were then tested in coverage-then-fill retrieval variants. A feature combination was kept only if the selected graph preserved the candidate order, Gold@20, candidate MRR@20, and candidate coverage while reducing the average evidence package or lowering shortcut-heavy evidence. On the PrimeKG validation audit, the selected Fuzzy-compact setting preserved the soft-support ranking metrics and candidate coverage while reducing the average subgraph from 60.45 to 32.56 triples. The intermediate Fuzzy-balanced and Fuzzy-shortcut variants retained 36.714 triples on average. All fuzzy variants reduced direct-shortcut exposure. This is why fuzzy retrieval is treated as evidence packaging rather than as a second candidate reranker.

S2.2 Weight Assignment Method

Fuzzy edge weights were fixed on validation data after the input candidate order had already been selected. Eligible settings had to preserve Gold@20 and candidate MRR@20, maintain candidate coverage, and reduce or clean the evidence graph. Candidate rank bands are assigned after the top-20 list is fixed: ranks 1–5 receive weight 1.00, ranks 6–10 receive weight 0.60, and ranks 11–20 receive weight 0.25. If an edge touches multiple candidates, Eq. (10) uses the highest touched-candidate band weight; if it touches no candidate and is not part of a candidate-anchored path, its band score is zero.

The density term in Eq. (13) uses $\eta_\rho = 10$ for PrimeKG, Hetionet, DRKG, and repoDB. PrimeKG uses a capped local-density hint, whereas Hetionet, DRKG, and repoDB use the log-degree density term shown above. PharmKG keeps its native local touch-count density hint. The density coefficient is 0.05 in the selected fuzzy rows, so density acts as a tie-breaking signal after candidate coverage, relation-family evidence, and penalty terms have been applied.

S2.3 Frozen Assigned Values

Table S2 reports the frozen edge-scoring weights.

$$\text{Eq. (10):} \quad B(e,q) = \max_{c_i \in C_q^{\text{in}} : e \sim c_i} w_b(r_i),$$

$$w_b(r_i) = \begin{cases} 1.00, & 1 \le r_i \le 5, \\ 0.60, & 6 \le r_i \le 10, \\ 0.25, & 11 \le r_i \le 20, \end{cases}$$

$$\begin{aligned} \text{Eq. (11):} \quad S_{\text{edge}}(e,q) &= \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\ &\quad + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\ &\quad - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned}$$

$$\begin{aligned} \text{Eq. (12):} \quad T_e^c(q) &= \mathbb{I}[e \text{ touches at least one candidate}], \\ T_e^q(q) &= \mathbb{I}[e \text{ touches the query entity}], \\ H_e(q) &= \mathbb{I}[e \text{ touches a top-band candidate}], \end{aligned}$$

$$\text{Eq. (13):} \quad \rho_e(q) = \frac{\log(1 + \deg_{S_q^{(0)}}(u)) + \log(1 + \deg_{S_q^{(0)}}(v))}{\eta_\rho}.$$

Table S2 Frozen fuzzy edge-scoring weights. These values instantiate Eq. (10), Eq. (11), Eq. (12), and Eq. (13). All datasets use rank bands 1–5, 6–10, and 11–20 with weights 1.0, 0.6, and 0.25, respectively.

Dataset	Candidate, query, and density weights	Relation-family priorities	Penalty terms
PrimeKG	In Eq. (10)–Eq. (11), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.05$, $\lambda_{\text{top}} = 0$, and $\lambda_\rho = 0.05$. In Eq. (13), density uses a capped local-density hint with $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 0$; no relation-family priority is used.	In Eq. (11), the direct shortcut penalty is $\lambda_d = 0.80$, and the contradiction penalty is $\lambda_x = 0.10$.
PharmKG	In Eq. (10)–Eq. (11), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.05$, $\lambda_{\text{top}} = 0$, and $\lambda_\rho = 0.05$. The density term keeps the native local touch-count hint.	In Eq. (11), $\lambda_f = 0$; no relation-family priority is used.	In Eq. (11), the direct shortcut penalty is $\lambda_d = 0.80$, and the contradiction penalty is $\lambda_x = 0$.
Hetionet	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is compound-gene 0.30, disease-gene 0.30, gene-gene 0.20, direct drug-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.50$ and $\lambda_x = 0$.
DRKG	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is compound-gene 0.30, disease-gene 0.30, gene-gene 0.20, target-treats 0.05, auxiliary compound-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.50$ and $\lambda_x = 0$.
repoDB	In Eq. (10)–Eq. (13), $\lambda_b = 1.0$, $\lambda_c = 0.35$, $\lambda_q = 0.10$, $\lambda_{\text{top}} = 0.05$, $\lambda_\rho = 0.05$, and $\eta_\rho = 10$.	In Eq. (11), $\lambda_f = 1.0$; $F_r(e)$ is target-approved 0.10, failed-diagnostic 0.02, compound-gene 0.35, gene-gene 0.25, other compound-disease 0.05, or other 0.05.	In Eq. (11), $\lambda_d = 0.40$ and $\lambda_x = 0.15$ for failed-diagnostic evidence.

S3 Fuzzy Retrieval Budgets

$$S_q^{\text{fuzzy}} = \text{TopEdges}\left(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\right), \quad \text{Eq. (14)}$$
$$C_q^{\text{fuzzy}} = C_q^{\text{in}}. \quad \text{Eq. (15)}$$

The budget values in Table S3 instantiate Eq. (14), while Eq. (15) fixes the candidate order after fuzzy graph selection.

Table S3 Frozen fuzzy retrieval budgets and coverage checks. These settings instantiate the graph budget $\tau_{d,q}$ in Eq. (14); candidate rankings are preserved as in Eq. (15).

Dataset	Budget rule	Selection passes	Frozen coverage result
PrimeKG	In Eq. (14), retain ratio 0.55, minimum 18, maximum equal to source size.	Candidate coverage pass, then global fill by edge score.	Validation 60.45 → 32.56 triples; test 59.93 → 32.34 triples. Candidate coverage preserved: 1.0.
PharmKG	In Eq. (14), retain ratio 0.55, minimum 18, maximum equal to source size.	Candidate coverage pass, then global fill by edge score.	Validation and test 100 → 55 triples. Candidate, top-band, middle-band, and tail-band coverage preserved: 1.0.
Hetionet	In Eq. (14), retain ratio 0.55, minimum 28, maximum 48.	Candidate coverage pass, query coverage pass, then global fill.	Validation and test 80 → 44 triples. Candidate coverage preserved: 1.0.
DRKG	In Eq. (14), retain ratio 0.55, minimum 45, maximum 60, family minimum 3.	Candidate coverage pass, query coverage pass, family preservation pass, then global fill.	Validation and test 100 → 55 triples. Candidate coverage preserved: 1.0.
repoDB	In Eq. (14), retain ratio 0.55, minimum 45, maximum 60. Family minimums are compound-gene 8, gene-gene 6, target 1, and failed 0.	Candidate coverage pass, query coverage pass, family preservation pass, then global fill.	Validation and test 100 → 55 triples. Candidate coverage preserved: 1.0.

S4 Algorithm

$$S_{\theta_d}(c_i, q) = \mu_d R_i(q) + (1-\mu_d) N_q(\widetilde{S}_{\theta_d}(c_i, q)), \quad \text{Eq. (8)}$$
$$c_i \prec c_j \iff S_i > S_j \text{ or } (S_i = S_j \text{ and } r_i^{(0)} < r_j^{(0)}), \quad \text{Eq. (9)}$$
$$\begin{aligned} S_{\text{edge}}(e, q) = {} & \lambda_f F_r(e) + \lambda_b B(e, q), \\ & + \lambda_c T_e^c(q) + \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) \\ & - \lambda_d D_e(q) - \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned} \quad \text{Eq. (11)}$$
$$S_q^{\text{fuzzy}} = \text{TopEdges}\left(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\right), \quad \text{Eq. (14)}$$
$$\mathbf{z}_e = \text{Adapter}\left([\mathbf{h}_e^{\text{global}}; \mathbf{h}_e^{\text{local}}]\right). \quad \text{Eq. (16)}$$

Algorithm S1 summarizes the complete inference sequence for one query.

Algorithm S1 SoftFuse-KGC inference for one query

Require: Query $q = (?, r_q, t_q)$, candidate universe $\mathcal{C}_d$, graph-compatible ranker, source graph retriever, support features, edge score S_{edge}

Retrieve the initial top-20 candidate list $C_q^{(0)}$.
Build the source evidence subgraph $S_q^{(0)}$ from the query, candidates, and available rule/path evidence.
Compute candidate support scores $S_{\theta_d}(c_i, q)$ using Eq. (8).
Re-rank candidates with the tie-breaking rule in Eq. (9) to obtain C_q^{soft}.
Assign candidate bands from C_q^{soft} and score evidence triples with Eq. (11).
Select S_q^{fuzzy} using the coverage and fill passes in Table S3.
Compute graph-enhanced query and candidate embeddings with the graph adapter in Eq. (16).
Insert `[QUERY]` and `[ENTITY]` embeddings into the candidate-constrained LLM prompt.
Generate the final answer from the calibrated candidate list.
return Prediction $\hat{y}_q$.

S5 Dataset Implementation Details

The end-to-end implementation uses a reproduced DrKGC-style graph-LLM setup: a graph-compatible R-GCN source provides candidate identifiers, local subgraphs, and structural embeddings for the graph adapter; the LLM is fine-tuned with LoRA; validation and test use $K = 20$ candidates. The primary dataset runs use Llama-3.2-3B with LoRA $r = 32$, $\alpha = 32$, dropout 0.1, learning rate 2×10^{-4}, source/target lengths $768/64$, deterministic one-beam decoding without sampling, and a single NVIDIA RTX 4090 GPU with 24GB memory. Table S4 reports the dataset-specific graph sources and frozen generation settings. These graph-adapter pipeline sources are distinct from the separately trained structure-only baselines reported later; their embedding dimensions therefore need not match.

Table S4 Graph-LLM implementation and decoding settings. Common LoRA, decoding, and hardware settings are stated in the paragraph above.

Dataset	Graph source	Training detail	Frozen decoding and graph size
PrimeKG	R-GCN; 4 rel.; 10453×256 embeddings.	Batch 1; grad. acc. 8.	New 16; rep. 1.00; no-repeat 0; graph $59.93 \rightarrow 32.34$.
PharmKG	R-GCN; 28 rel.; 7247×128 embeddings.	Batch 1; grad. acc. 16.	New 16; rep. 1.05; no-repeat 0; graph $100 \rightarrow 55$.
Hetionet	R-GCN-compatible; 24 rel.; 47031×128 embeddings.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $80 \rightarrow 44$.
DRKG	R-GCN consistency source; 85 rel.; R-GCN-ready IDs.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $100 \rightarrow 55$.
repoDB	R-GCN; 63 rel.; readable names plus canonical IDs.	Batch 1; grad. acc. 8.	New 16; one beam; no sampling; graph $100 \rightarrow 55$.

Table S5 lists the question templates and evidence-path retrieval packages used for each dataset.

Table S5 Pre-specified question templates and evidence-path retrieval rules. Task templates and retrieval rules format candidate-constrained prompts and package local evidence.

Dataset	Question template	Path-rule or retrieval package
PrimeKG	*What drug is indicated for {disease}?*	Drug–target–gene–disease and drug–target–gene–ppi–gene–disease paths; contraindication-like drug–disease evidence is retained as a negative/safety cue.
PharmKG	*What drug is therapeutically associated with {disease}?*	The T relation is the therapeutic-association proxy; retrieval uses incident-edge and shortest-path fallback evidence.
Hetionet	*What compound treats {disease}?*	CbG–DaG, CdG–DdG, CuG–DuG, one-hop GiG bridges, and direct CpD evidence.
DRKG	*What compound treats {disease}?*	Compound:Gene–Gene:Disease, direct Compound:Disease, and Gene:Gene bridge evidence for the DrugBank treats head-prediction target.
repoDB	*What drug is approved for {disease}?*	Compound:Gene–Gene:Gene, Compound:Disease, and failed-diagnostic evidence retained as diagnostic rather than positive approved-indication context.

S5.1 Prompt and Output Example

All datasets use the same candidate-constrained prompt pattern. The following PrimeKG example illustrates the input schema and supervised output:

> You are a biomedical scientist. The task is to predict the answer based on the given question, and you only need to answer one entity. The answer must be in ('Fusidic acid', 'Cortisone acetate', 'Hydrocortisone', 'Betamethasone', 'Tetracycline', 'Ofloxacin', 'Ciprofloxacin', 'Demeclocycline', 'Dexamethasone', 'Triamcinolone', 'Prednisolone', 'Methylprednisolone', 'Hydrocortisone acetate', 'Methdilazine', 'Norfloxacin', 'Doxycycline', 'Prednisone', 'Alimemazine', 'Doxylamine', 'Phenylephrine').
> You can refer to the entity embeddings: 'allergic rhinitis': `[QUERY]`, 'Fusidic acid': `[ENTITY]`, 'Cortisone acetate': `[ENTITY]`, 'Hydrocortisone': `[ENTITY]`, ..., 'Phenylephrine': `[ENTITY]`.
> Question: What drug is indicated for allergic rhinitis?
> Answer:

The corresponding supervised output is *Hydrocortisone*. During evaluation, the generated string is normalized and matched against the supplied candidate list; unmatched strings are counted as invalid predictions.

S6 PrimeKG Variant Formulas

PrimeKG is used here as the representative formula summary.

S6.1 Soft-Support Variants

$$
\begin{aligned}
\text{Selected family:}\quad & \widetilde{S}(c_i, q) = \alpha_E E_i(q) - \alpha_D D_i(q) - \alpha_X X_i(q), \\
& \text{b025}: \quad E_i(q) - 0.25 D_i(q) - 0.05 X_i(q), \\
& \text{b050}: \quad E_i(q) - 0.50 D_i(q) - 0.10 X_i(q), \\
& \text{bcap}: \quad \min(E_i^{\text{count}}(q), 2) - 0.50 D_i(q) - 0.10 X_i(q).
\end{aligned}
$$

The selected PrimeKG contact-support family is a special case of Eq. (8) with $\mu = 0$. The final row uses b050. The b025 row tied b050 in validation, but b050 was retained because it applies the stronger shortcut penalty while preserving the same candidate ranking. The capped-evidence contrast underperformed because it reintroduced shortcut-heavy candidates.

S6.2 Fuzzy-Retrieval Variants

$$S_{\text{edge}}(e,q) = B(e,q) + 0.35T_e^c(q) + 0.05T_e^q(q) - \lambda_d D_e(q) - 0.10X_e(q) + 0.05\rho_e(q).$$

The PrimeKG fuzzy edge score uses Eq. (11) with no relation-family term. The selected Fuzzy-compact variant (called *tight* in the configuration audit) sets $\lambda_d = 0.80$, retain ratio 0.55, and minimum keep 18. The Fuzzy-shortcut contrast (called *directplus* in the audit) sets $\lambda_d = 1.00$, retain ratio 0.67, and minimum keep 20. Both preserve candidate ranking, but Fuzzy-compact gives the smaller and cleaner graph package on validation. Table S6 summarizes the selected and contrastive PrimeKG rows.

Table S6 PrimeKG variant-selection summary. Candidate metrics are validation metrics for soft support and held-out test metrics for the locked ablation rows.

Row	Gold@20	MRR@20	Hits@1	Hits@10	Selection note
Backbone raw, validation	0.202	0.0918	0.024	0.158	Reference candidate order.
Ontology hard filter, validation	0.014	0.0509	0.002	0.012	Negative control; brittle under missing evidence.
Soft support b050, validation	0.202	0.1356	0.056	0.182	Selected soft-support row.
Capped-evidence contrast, validation	0.202	0.0925	0.022	0.158	Retained only as a contrast row.
Backbone raw, test	0.240	0.0646	0.024	0.192	Locked test reference.
Soft support, test	0.240	0.1253	0.072	0.222	Candidate-stage gain.
Fuzzy retrieval, test	0.240	0.1253	0.072	0.222	Candidate order preserved; graph $59.93 \rightarrow 32.34$.

S7 LLM Model Comparison

The model comparisons use candidate-constrained evaluation: generated strings are normalized against the supplied candidate list, and invalid outputs receive zero reciprocal rank. Table S7 reports the PrimeKG generator comparison.

Table S7 PrimeKG LLM comparison on the fuzzy retrieval row. All rows use the same candidate list and fuzzy evidence package. The configuration column reports max new tokens / repetition penalty / no-repeat n-gram size.

Model	Config	E2E MRR@20	E2E Hits@10	Pred. in cand.	Invalid	Top1 copy	Interpretation
Llama-3.2-3B	16/1.00/0	0.0747	0.218	0.998	0.002	0.010	Primary generator; high candidate validity without top-ranked candidate copying.
Llama-3-8B	16/1.00/0	0.1253	0.222	1.000	0.000	1.000	Diagnostic generator behavior; all test outputs follow the first candidate.
MedLlama-3-8B	16/1.00/0	0.1253	0.222	1.000	0.000	1.000	Same candidate-order-following behavior as Llama-3-8B.

Table S8 reports the corresponding PharmKG comparison.

Table S8 PharmKG LLM comparison on the fuzzy retrieval row. The first three rows use the frozen PharmKG decoding configuration. Mistral-7B is included as a diagnostic comparison.

Model	Config	E2E MRR@20	E2E Hits@10	Pred. in cand.	Invalid	Top1 copy	Interpretation
Llama-3.2-3B	16/1.05/0	0.0210	0.062	0.412	0.588	0.412	Most stable PharmKG generator among the frozen rows.
Llama-3-8B	16/1.05/0	0.0203	0.064	0.274	0.726	0.274	Lower candidate validity; no PrimeKG-like top1-copy collapse.
MedLlama-3-8B	16/1.05/0	0.0186	0.060	0.022	0.978	0.022	Mostly invalid under the candidate-constrained output rule.
Mistral-7B	32/1.00/0	0.0186	0.060	0.014	0.986	0.000	Diagnostic comparison; candidate validity remains very low.

Table S8 shows a different failure mode from the PrimeKG 8B rows. Llama-3.2-3B gives the highest candidate-constrained validity on PharmKG, although invalid outputs remain frequent because the transfer task uses a noisier therapeutic-association proxy. Llama-3-8B has slightly higher Hits@10 but lower candidate validity, while MedLlama-3-8B and Mistral-7B mostly generate strings outside the supplied candidate list.

S7.1 Decoding-Configuration Check

The deterministic decoding sweep used for configuration selection varied max new tokens, repetition penalty, and no-repeat n-gram size. The sweep was performed on the validation split with the primary generator, and the selected configuration was then held fixed for the model comparison. This avoids selecting a separate decoding policy for each larger model after observing comparison outcomes. Table S9 therefore separates two checks: the full eight-configuration sweep showing that the frozen decoding setting itself does not induce top1-copy behavior in Llama-3.2-3B, and the larger-model frozen checks showing that top1-copy appears when the generator is changed to Llama-3-8B or MedLlama-3-8B under the same selected configuration.

Table S9 PrimeKG decoding-configuration check for top1-copy behavior. Top1-copy means that the generated entity is the first candidate in the supplied list. The Llama-3.2-3B rows report the complete validation sweep; the 8B rows report the frozen configuration used in the model comparison.

Model/protocol	Config	New	Rep.	No-rep.	Val. E2E MRR@20 / Hits@10	Val. top1 copy	Test top1 copy	Interpretation
Llama-3.2-3B sweep	cfg01	16	1.00	0	0.0585/0.174	0.008	0.010	Frozen setting selected because all swept settings tied on the fuzzy row.
Llama-3.2-3B sweep	cfg02	16	1.05	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg03	16	1.10	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg04	16	1.10	3	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg05	24	1.00	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg06	24	1.05	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg07	24	1.10	0	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3.2-3B sweep	cfg08	24	1.10	3	0.0585/0.174	0.008	–	Same validation metrics as cfg01.
Llama-3-8B frozen check	cfg01	16	1.00	0	0.0976/0.182	1.000	1.000	Test outputs: 464 non-gold top1 copies and 36 exact-target top1 cases.
MedLlama-3-8B frozen check	cfg01	16	1.00	0	0.0976/0.182	1.000	1.000	Same validation and test breakdown as Llama-3-8B.

The PrimeKG sweep shows that changing deterministic decoding length, repetition penalty, and no-repeat n-gram size did not change the primary generator's fuzzy-row validation metrics or create top1-copy collapse. The collapse appears only in the larger-model frozen checks, where both 8B generators copy the first candidate for every validation and test example. Their higher adjusted E2E MRR@20 is therefore interpreted as candidate-order following after the soft-support/fuzzy candidate list has already been supplied, not as stronger use of the evidence text or graph package.

S8 Ablation, Sensitivity, and Robustness

Table S10 isolates the candidate-reranking and evidence-packaging contributions on the PrimeKG test split.

Table S10 PrimeKG held-out test ablation. The soft-support row changes candidate ordering; fuzzy retrieval changes evidence packaging while preserving the candidate order.

Row	Gold@20	MRR@20	Hits@1	Hits@10	Avg. graph	Note
Backbone raw	0.240	0.0646	0.024	0.192	59.93	Initial graph-compatible source.
Soft support	0.240	0.1253	0.072	0.222	59.93	Candidate $\Delta\mathrm{MRR@20} = +0.0608$.
Fuzzy retrieval	0.240	0.1253	0.072	0.222	32.34	Ranking unchanged; graph reduced by 27.59 triples on average.

Table S11 evaluates retrieval-rule and question-template sensitivity with a fixed candidate order.

Table S11 PrimeKG rule-package and question-template sensitivity. Candidate order is fixed in these checks.

Variant	Candidate MRR@20	E2E MRR@20	E2E Hits@10	Invalid	Avg. graph	Note
Main rules	0.1253	0.0747	0.218	0.002	32.34	Canonical setting.
No rules	0.1253	0.0747	0.218	0.020	59.93	Similar rank metric but larger graph and higher invalid rate.
Random rules	0.1253	0.0747	0.218	0.002	32.34	Diagnostic control; not selected over main rules.
Canonical question	0.1253	0.0747	0.218	0.002	32.34	Baseline wording.
Treatment wording	0.1253	0.0747	0.218	0.000	32.34	Prediction string changes in 3.0% of test examples.
Medication wording	0.1253	0.0747	0.218	0.002	32.34	Prediction string changes in 0.6% of test examples.
Association-neutral wording	0.1253	0.0747	0.218	0.000	32.34	Prediction string changes in 4.6% of test examples.

Table S12 reports small score perturbations and edge-drop robustness checks.

Table S12 PrimeKG small-noise robustness on the held-out test split. Score-noise variants perturb support scores before reranking; edge-drop variants remove approximately 5% of selected subgraph edges.

Variant	MRR@20	Hits@10	Same top1	Rank change	Avg. graph
No noise	0.1253	0.222	1.000	0.000	32.34
Score noise s1	0.0779	0.220	0.086	0.206	32.34
Score noise s2	0.0779	0.216	0.096	0.214	32.34
Score noise s3	0.0763	0.220	0.086	0.214	32.34
Edge drop s1	0.1253	0.222	1.000	0.000	30.49
Edge drop s2	0.1253	0.222	1.000	0.000	30.49
Edge drop s3	0.1253	0.222	1.000	0.000	30.49

The sensitivity checks show that the PrimeKG result is not driven by one question wording. The robustness check separates two effects: the selected fuzzy graph is stable under light edge dropout, whereas support-score perturbation changes close candidate-score ties and reduces MRR@20. This highlights a calibration limitation of the current support score.

S9 Fuzzy Encoder Probe

The fuzzy encoder probe tests whether fuzzy evidence weights should enter message passing directly rather than only the retrieval stage. It remains diagnostic because it worsens more validation cases than it improves. Table S13 reports the validation comparison.

Table S13 PrimeKG fuzzy encoder probe on validation. The probe uses the same validation split as the retrieval-selection comparison.

Row	Gold@20	MRR@20	Hits@1	Hits@3	Hits@10	Probe note
Soft support	0.202	0.1356	0.056	0.134	0.182	Candidate-stage reference.
Fuzzy retrieval	0.202	0.1356	0.056	0.134	0.182	Candidate order preserved; avg. graph 32.56.
Fuzzy encoder probe	0.202	0.0917	0.012	0.058	0.176	Improves 8 queries, worsens 70, and leaves 422 unchanged relative to fuzzy retrieval.

The probe confirms that fuzzy scores are useful for selecting an evidence package, but the current weighted-message-passing formulation is not yet stable enough to replace the selected retrieval-stage design.

S10 Structure-Only Baseline Training Settings

Table S14 reports the training settings for all structure-only baselines.

Table S14 Structure-only baseline training settings. Translation/bilinear baselines include TransE, DistMult, ComplEx, and RotatE. GNN baselines include R-GCN and HRGAT.

Dataset	Translation and bilinear baselines	GNN baselines	Candidate and graph protocol
PrimeKG	TransE, DistMult, ComplEx, RotatE: dimension 128, 80 epochs, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	R-GCN and HRGAT: dimension 128, 40 epochs, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	Drug-only universe: 1,801 candidates; training graph: 136,351 triples; target train pairs: 8,388.
PharmKG	TransE, DistMult, ComplEx, RotatE: dimension 256, 30 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2025.	R-GCN and HRGAT: dimension 128, two graph layers, 40 epochs, 32,768 sampled edges per epoch, dropout 0.1, learning rate 10^{-3}, weight decay 10^{-5}, seed 2025.	Training heads of relation T: 1,342 candidates; training graph: 386,768 triples and 28 relations.
Hetionet	TransE, DistMult, ComplEx, RotatE: dimension 128, 20 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2026.	R-GCN: dimension 128, two graph layers, 80 epochs. HRGAT: dimension 64, two graph layers, 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Compound universe: 1,552 candidates; training graph: 2,249,997 edges and 24 relations.
DRKG	TransE, DistMult, ComplEx, RotatE: dimension 128, 30 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2027.	R-GCN and HRGAT: dimension 128, two graph layers; R-GCN uses 80 epochs and HRGAT uses 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Training target heads: 1,542 candidates; training graph: 187,962 edges and 85 relations.
repoDB	TransE, DistMult, ComplEx, RotatE: dimension 128, 40 epochs, batch size 4096, learning rate 10^{-3}, weight decay 10^{-6}, margin 1.0, seed 2028.	R-GCN and HRGAT: dimension 128, two graph layers; R-GCN uses 80 epochs and HRGAT uses 60 epochs. Both use target/support samples of $8192/8192$, support loss weight 0.25, dropout 0.1, learning rate 10^{-3}, and weight decay 10^{-5}.	Approved training heads: 1,519 candidates; training graph: 90,804 edges and 63 relations.

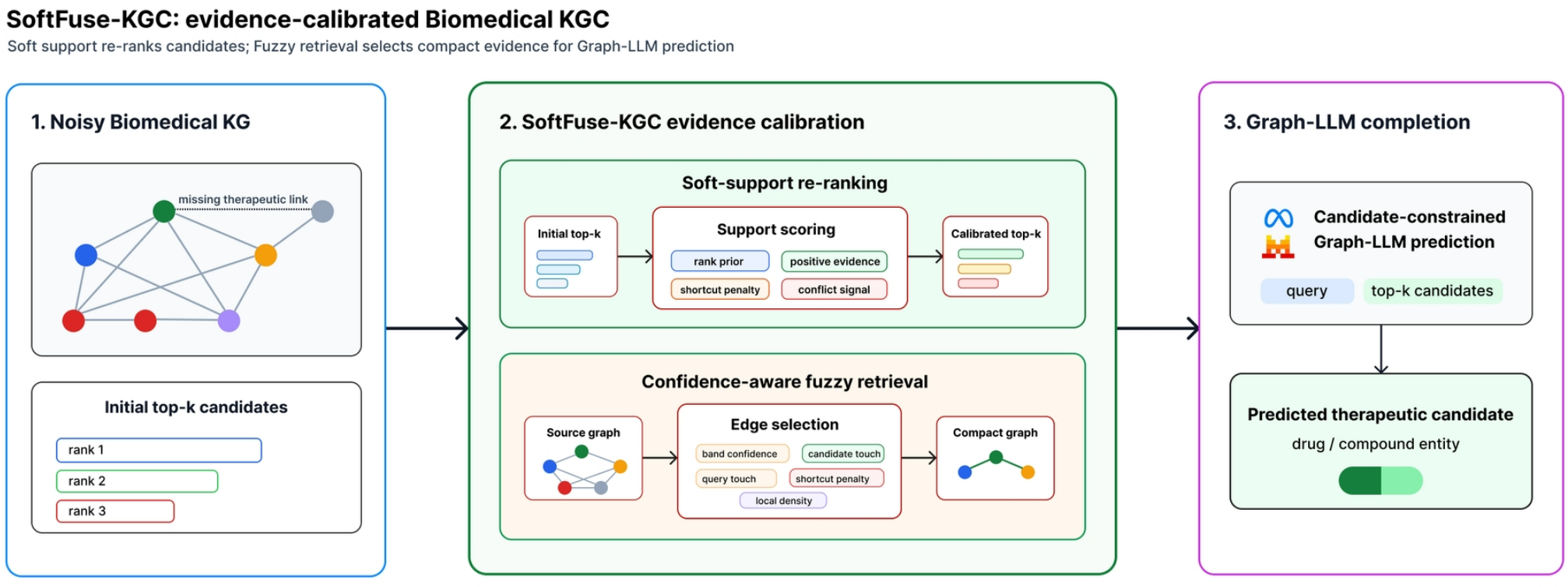
S11 Dataset Preprocessing and Leakage Checks

$$q = (?, r_q, t_q), \qquad (c_q^\star, r_q, t_q) \notin \mathcal{G}_{\text{evidence}}^{\text{train}} \quad \text{for validation and test queries.}$$

All datasets are converted into head-prediction tasks of the form $(?, r_q, \text{disease})$. Validation and test splits are constructed before candidate scoring, and the held-out positive target triples are removed from the training evidence graph. Table S15 summarizes target construction, evidence-graph preparation, and leakage checks.

Table S15 Dataset preprocessing and leakage checks. Coverage means that validation/test gold drugs and query diseases remain represented in the training candidate/query universe. Exact leakage means that held-out target triples are absent from the training evidence graph and prepared evaluation inputs.

Dataset	Target construction and split	Evidence graph construction	Leakage and protocol checks
PrimeKG	Drug-restricted indication head prediction with $8{,}388/500/500$ train/validation/test rows. Candidate universe is drug-only, and $K = 20$ candidates are supplied without validation/test gold injection.	Four-relation enriched graph: indication, target, associated-with, and protein-protein interaction. Evaluation prompts align candidates and evidence by query; supervised targets appear only in training outputs.	Validation/test exact-leak counts are 0. Training rows contain supervised targets by design; validation/test rows use no-injection candidate lists.
PharmKG	Relation T is a therapeutic-association proxy. Coverage filtering, train-overlap removal, forbidden-pair removal, and deduplication give $28{,}960/500/500$ rows.	Training T-heads define $1{,}342$ candidates. The enriched graph has $386{,}768$ triples, $7{,}247$ entities, and 28 relations.	Leakage checks pass: valid-positive-in-train $= 0$, test-positive-in-train $= 0$, valid-test overlap $= 0$, enriched-graph held-out target count $= 0$; type errors $= 0$.
Hetionet	The target relation is CtD, with $555/100/100$ train/validation/test rows. The split enforces that validation/test compounds and diseases remain in the training CtD relation.	Held-out CtD target edges are removed from the enriched graph, which contains $2{,}249{,}997$ training edges over $47{,}031$ entities and 24 relations.	Coverage passes for validation and test. The 200 held-out target edges are removed, and exact-leak count is 0 in the graph, raw evaluation inputs, soft-support inputs, and fuzzy-retrieval inputs.
DRKG	The target relation is DrugBank treats Compound-to-Disease, with $3{,}968/500/500$ train/validation/test rows. Validation/test compounds and diseases remain in the training target-relation universe.	The enriched graph keeps selected compound-gene, target-treats, auxiliary compound-disease, disease-gene, and gene-gene families, yielding $187{,}962$ edges over $13{,}918$ entities and 85 relations.	Coverage passes for validation and test. The $1{,}000$ held-out target edges are skipped during graph construction, and valid/test exact-leak counts are 0 in the graph and prepared evaluation inputs.
repoDB	Approved indication pairs form the target relation. Failed or suspended pairs are retained only as diagnostic evidence after conflict handling. The approved target split is $5{,}677/500/500$.	Diseases are local UMLS nodes; mapped DrugBank compounds reuse DRKG-derived compound-gene and gene-gene evidence. The graph has $90{,}804$ edges, $8{,}818$ entities, and 63 relations.	Coverage and schema checks pass. Validation and test exact-leak counts are 0; all train, validation, and test entities are mapped into the prepared graph.


SoftFuse-KGC: evidence-calibrated Biomedical KGC
Soft support re-ranks candidates; Fuzzy retrieval selects compact evidence for Graph-LLM prediction
1. Noisy Biomedical KG
missing therapeutic link
Initial top-k candidates
rank 1
rank 2
rank 3
2. SoftFuse-KGC evidence calibration
Soft-support re-ranking
Initial top-k
Support scoring
rank prior
positive evidence
shortcut penalty
conflict signal
Calibrated top-k
Confidence-aware fuzzy retrieval
Source graph
Edge selection
band confidence
candidate touch
query touch
shortcut penalty
local density
Compact graph
3. Graph-LLM completion
Candidate-constrained
Graph-LLM prediction
query
top-k candidates
Predicted therapeutic candidate
drug / compound entity

SoftFuse-KGC: Soft Support and Confidence-Aware Fuzzy Retrieval for Biomedical Knowledge Graph Completion

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Abstract

Biomedical knowledge graph completion can support drug repurposing, but graph-augmented large language model (LLM) pipelines are sensitive to poorly ordered candidates and noisy query-specific evidence. In this paper, we introduce SoftFuse-KGC, an evidence-calibration framework for drug-focused link prediction with graph-augmented LLMs. Soft-support module of the proposed model reranks top-20 candidates using graded evidence contact, mechanism-oriented support, and penalties for direct shortcuts and contradictory signals. Its confidence-aware fuzzy retriever selects a compact evidence graph using rank confidence, query and candidate incidence, penalty signals, and local density while preserving candidate order. We evaluate the framework on PrimeKG, PharmKG, Hetionet, the Drug Repurposing Knowledge Graph (DRKG), and repoDB. On PrimeKG, SoftFuse-KGC raises candidate mean reciprocal rank at 20 (MRR@20) from 0.0646 to 0.1253 and end-to-end MRR@20 from 0.0476 to 0.0747 while reducing the average graph from 59.93 to 32.34 triples. On PharmKG, it raises end-to-end MRR@20 from 0.0156 to 0.0210 and reduces the graph from 100 to 55 triples. On Hetionet, soft support improves end-to-end MRR@20 from 0.1045 to 0.1109. On DRKG and repoDB, fuzzy retrieval improves end-to-end MRR@20 while reducing average graph size by 45%. These results show that candidate calibration and evidence compression offer complementary benefits whose effectiveness depends on graph topology and query-specific support. Furthermore, to validate the quality of the proposed model, we present two case studies across three diagnostic groups from PrimeKG. Additionally, a web interface has been developed to facilitate the inspection of candidate support scores, ranking changes, and evidence-containing subgraphs retrieved via a fuzzy mechanism.

Keywords: Biomedical knowledge graph · Knowledge graph completion · Drug repurposing · Soft evidence modeling · Fuzzy retrieval

1 Introduction

Knowledge graphs (KGs) represent entities and their typed relations in a form that supports integrating, analyzing, and reasoning over heterogeneous knowledge [1–3]. This representation is particularly useful in biomedicine, where relevant evidence is distributed across drugs, diseases, genes, proteins, pathways, adverse events, phenotypes, and molecular mechanisms. Biomedical knowledge graphs (BKGs) and related resources, including Hetionet [4], PrimeKG [5], PharmKG [6], DRKG [7], repoDB [8], and others organize this evidence into computable graph structures for translational discovery, drug repurposing,

safety modeling, and therapeutic hypothesis generation. These resources integrate biomedical terminologies and molecular databases such as UMLS [9], DisGeNET [10], UniProt [11], OMIM [12], DrugBank [13], STITCH [14], and STRING [15]. Despite their broad coverage, curated BKGs remain incomplete. Biomedical facts may be missing from databases, dispersed across publications, represented with inconsistent identifiers, or not yet known.

Knowledge graph completion (KGC), often formulated as link prediction, addresses this incompleteness by ranking plausible missing triples from the observed graph [1, 2, 16]. Many established KGC methods rely primarily on graph structure. Translation and semantic matching models such as TransE [17], DistMult [18], ComplEx [19], and RotatE [20] learn representations of entities and relations and use them to score candidate triples efficiently. Graph neural networks such as R-GCN [21] and CompGCN [22], together with relation-aware graph attention models [23], extend this approach by aggregating information from multirelational neighborhoods, allowing local graph context and multi-hop biomedical evidence to influence prediction. Rule-based and path-based methods provide more interpretable predictions through relational patterns and mechanistic paths. However, these methods can become brittle when biomedical evidence is sparse or when the graph does not contain a useful path [24–26].

Textual information provides another source of evidence for biomedical KGC. Entity names, relation labels, and biomedical descriptions contain semantic information that structure-only models may not capture. Language-model-based KGC and text-enhanced ranking methods therefore use relation descriptions, entity names, instruction prompts, retrieval, or candidate-constrained generation to connect graph prediction with biomedical language [27–34]. More recent graph-augmented LLM systems combine candidate retrieval, graph evidence retrieval, graph adapters, and parameter-efficient fine-tuning so that generation can be conditioned more directly on retrieved graph evidence. DrKGC [35] represents this direction by integrating question templates, lightweight candidate retrieval, rule-guided subgraph retrieval, graph-adapter embeddings, and LoRA [36] fine-tuning. For biomedical applications, this design provides a language-based prediction interface while retaining structured evidence from the graph.

The connection between retrieval and generation nevertheless creates two important bottlenecks. First, a candidate-constrained LLM can select only from the entities provided by the candidate retriever. If the gold drug is absent from the top-K candidate list, the generation stage cannot recover it. Even when the gold drug is present, a position near the end of the list may limit its influence on generation and make the prediction less useful for manual review. Second, the graph adapter can use only the evidence contained in the retrieved subgraph. A large or noisy graph may be dominated by direct shortcuts, frequently connected entities, or relations that are structurally close to the query but offer limited clinical or mechanistic information. These limitations are especially relevant to therapeutic head prediction, in which the system ranks drugs or compounds for an incomplete triple of the form $(?, r, \text{disease})$. Evidence for such a query may be indirect and may involve drug targets, disease-associated genes, protein interactions, pathways, previous indications, failed indications, or literature-derived associations.

These retrieval constraints motivate a graded treatment of biomedical evidence. A drug–target edge, disease–gene association, protein interaction, therapeutic association, approved indication, and contraindication-like signal provide different types and strengths of evidence. At the same time, the absence of a path in an incomplete BKG should not automatically be interpreted as negative evidence. Although hard filters are easy to inspect, they may remove recoverable candidates simply because the graph lacks a particular supporting edge. Fuzzy set theory introduced graded membership as an alternative to binary inclusion [37], and fuzzy logic subsequently provided formal tools for reasoning with partial truth and uncertainty [38–41]. Applied to biomedical graph-LLM completion, this perspective suggests that candidate support and graph evidence should be calibrated according to their degree of support, their susceptibility to shortcut behavior, and the confidence associated with the retrieved context.

We address this retrieval–generation interface with SoftFuse-KGC, an evidence-calibration framework for DrKGC-style biomedical graph-augmented LLM completion. SoftFuse-KGC contains two complementary modules. The soft support module reranks retrieved drug candidates using graded evidence contact, mechanism-oriented support, penalties for direct shortcuts, and contradiction cues. The confidence-aware fuzzy retrieval module selects a compact evidence subgraph using candidate rank bands, candidate and query incidence, relation-family priorities, local density, and penalty signals, while preserving the selected candidate order. SoftFuse-KGC is not intended to replace the upstream structural candidate generator. Instead, it evaluates three distinct aspects that are often conflated in graph-LLM pipelines: gold-candidate recoverability, the ordering of recoverable candidates, and the compactness and auditability of the evidence supplied to the graph adapter.

The main contributions of this paper are as follows:

- We introduce a soft support reranking module for drug-focused biomedical KGC. The module converts heterogeneous graph signals into graded candidate support instead of applying a binary keep-or-discard rule.
- We propose a confidence-aware fuzzy retrieval module that compresses query-specific evidence while preserving candidate coverage and the selected candidate order. The resulting graph-LLM inputs are smaller and easier to inspect.
- We evaluate the proposed model SoftFuse-KGC on five biomedical datasets—PrimeKG, PharmKG, Hetionet, DRKG, and repoDB—and report candidate coverage, ranking quality, end-to-end LLM generation, graph size, shortcut exposure, and case-level auditability.
- We present two case studies using the PrimeKG dataset across three diagnostic categories to validate the proposed model. Additionally, we also developed a website application that provides an interface for the interactive inspection of the proposed model SoftFuse-KGC, this application supports reproducible inspection of candidate supports scores, rank changes, and fuzzy-retrieved evidence subgraphs.

The remainder of this paper is organized as follows. Section 2 reviews structure-based biomedical KGC, graph-augmented LLMs, and soft evidence modeling. Section 3 presents the SoftFuse-KGC framework, including its graph-LLM backbone, soft support module, fuzzy retrieval module, and graph-adapter-enhanced generation. Section 4 describes the study datasets and evaluation metrics before reporting the candidate-stage, end-to-end, case-study, and web-server results. Section 5 discusses the findings and limitations. Section 6 concludes the paper and outlines future research directions.

2 Related Work

2.1 Structure-based biomedical KGC

Translation and semantic matching models such as TransE [17], DistMult [18], ComplEx [19], and RotatE [20] represent entities and relations in continuous embedding spaces and score candidate triples from these representations. Their efficient scoring functions make them useful as large-scale candidate generators, although their predictions are driven mainly by structural regularities in the observed graph. Graph neural networks such as R-GCN [21] and CompGCN [22], together with relation-aware graph attention variants [23], incorporate information from multirelational neighborhoods into entity representations. These models represent local relational context more explicitly than conventional embedding models. Rule-based and path-based approaches instead identify interpretable relational patterns or paths that connect candidate entities to a query. Their dependence on observable paths, however, can make them sensitive to missing evidence [24–26]. Biomedical link prediction and drug repurposing studies further demonstrate that graph proximity must be interpreted carefully because structural closeness alone does not establish therapeutic appropriateness [4, 42–44].

2.2 Text and graph-augmented LLMs

Text-enhanced and LLM-based KGC methods use relation descriptions, entity names, instruction tuning, retrieval, or candidate-constrained generation to complement structural graph information [27–34]. This approach is relevant to BKGs because biomedical entity names and relation labels contain domain-specific semantic information. Text-only prompting, however, may omit important graph structure, whereas unconstrained generation can produce entities that do not belong to the graph or the intended candidate set.

DrKGC [35] is particularly relevant to the present study because it combines question templates, lightweight candidate retrieval, rule-guided dynamic subgraph retrieval, graph-adapter embeddings, and LoRA [36] fine-tuning. Related biomedical graph-LLM research also shows the value of combining textual representations with domain-specific graph evidence [45]. Nevertheless, a DrKGC-style system remains dependent on two retrieved inputs: the ordered candidate list and the query-specific evidence graph supplied to the graph adapter. Errors in either input can propagate to candidate-constrained generation.

More broadly, retrieval-augmented generation treats external evidence as a non-parametric memory for knowledge-intensive tasks, allowing a generator to use retrieved context in addition to information stored in its parameters [46]. Surveys of KG–LLM integration similarly distinguish KG-enhanced LLMs, LLM-augmented KGs, and systems that combine the two components more closely [47]. In biomedical KGC, these research directions highlight a practical problem at the retrieval–generation interface: prioritizing retrieved candidates and allocating the limited evidence budget to the most relevant local triples.

2.3 Soft evidence for graph-LLM completion

The present work addresses this interface between a structural candidate source and the graph-LLM input package. Its two central tasks are to promote well-supported candidates within a fixed candidate set and to retain informative evidence triples for downstream graph-enhanced generation.

Soft evidence modeling is appropriate for this setting because biomedical graph relations are heterogeneous. A drug–target edge, disease–gene association, protein interaction, literature-derived drug–disease association, and approved indication do not provide equivalent therapeutic evidence. Hard filtering remains interpretable, but it can be unreliable in an incomplete BKG: The absence of an observed supporting relation does not necessarily indicate the absence of biomedical support. Fuzzy set theory represents membership as a degree rather than a binary decision [37], while fuzzy logic provides foundations for reasoning with partial truth and uncertainty [38–41]. Fuzzy knowledge graphs have also been combined with complex fuzzy inference systems to support approximate reasoning when the available knowledge or rule base is limited [48]. In clinical decision support, picture fuzzy sets and knowledge-based reasoning have been used to represent imprecise diagnostic information [49, 50]. SoftFuse-KGC applies the same general principle to evidence calibration in biomedical graph-LLM completion. Observed relations, missing support, indirect paths, shortcut signals, and contradiction-like evidence remain distinguishable instead of being collapsed prematurely into a single binary keep-or-discard decision.

3 Methods

3.1 DrKGC-style graph-LLM backbone

The backbone retains the central components of DrKGC [35]: candidate retrieval, source-subgraph retrieval, graph-adapter embedding enhancement, and candidate-constrained LLM prompting. Across all datasets, we use a lightweight R-GCN [21] ranker as the graph-compatible source. This choice is motivated by interface compatibility rather than by an assumption that R-GCN is the strongest standalone link predictor. R-GCN provides relation-aware entity embeddings and local neighborhoods in the format expected by the DrKGC-style subgraph encoder. Using the same source also keeps the graph-adapter interface fixed and makes the calibration modules comparable across datasets.

The calibration modules themselves are not tied to R-GCN. Any structural or hybrid retriever can serve as the source if it provides a fixed candidate list, compatible entity identifiers or embeddings, and a query-specific subgraph. Integrating stronger compatible sources, such as ComplEx [19], DistMult [18], relation-aware attention retrievers [23], or hybrid retrievers, remains future work. The initial candidate list is defined as follows:

$$C_q^{(0)} = (c_1, c_2, \ldots, c_K), \quad K = 20, \tag{1}$$

where the backbone structural score determines the order. Candidate names are inserted explicitly into the prompt, which instructs the LLM to return one entity from this set. The source subgraph is intentionally larger than the final fuzzy graph. This design preserves the original DrKGC graph-adapter interface while retaining enough local evidence for SoftFuse-KGC to determine which candidate and edge signals should be retained or downweighted.

3.2 Overview of the SoftFuse-KGC framework

SoftFuse-KGC operates as an evidence-calibration layer between candidate retrieval and graph-adapter-enhanced generation. Fig. 1 shows the four implementation stages. **A. Biomedical backbone preparation** constructs the task-specific BKG, trains or loads the graph-compatible drug ranker, exports global entity embeddings, and prepares path rules and question templates. **B. Candidate and evidence retrieval** receives a therapeutic query of the form $(?, r_q, \text{disease})$, retrieves the top-20 drug candidates, and packages a source subgraph containing direct candidate–query edges, candidate neighborhoods, and rule-compatible local paths. **C. SoftFuse-KGC evidence calibration** applies the two proposed modules: C1 soft support reranks candidates from graded support and penalty cues, whereas C2 confidence-aware fuzzy retrieval selects a smaller evidence graph using rank-band confidence and edge-level evidence quality. **D. Graph-enhanced candidate-constrained generation** passes the calibrated candidates and selected graph to the graph adapter and LoRA-tuned LLM, which must return one supplied candidate.

The distinction between C1 and C2 is central to the study design. We attribute changes in candidate order to soft support and changes in graph size or evidence content to fuzzy retrieval.

3.3 Soft support candidate reranking

The soft support module reranks the backbone candidates using evidence extracted from the initial source subgraph $S_q^{(0)}$. This is the graph package constructed by the DrKGC-style retriever before fuzzy compression. It contains candidate–query paths, candidate-neighborhood edges, and relation-specific evidence that may provide support, indicate risk, or create an ambiguous connection between a candidate and the query. Soft support converts these heterogeneous signals into a calibrated candidate score rather than making a binary keep-or-discard decision.

We formulate soft support as a unified candidate score. It combines positive evidence $E_i(q)$, direct-shortcut or popularity evidence $D_i(q)$, contradiction-like evidence $X_i(q)$, and, where selected, a backbone rank prior. Dataset-specific instantiations set the coefficients of unavailable or inactive terms to zero.

For query q and candidate c_i, the positive support family is computed as follows:

$$\begin{aligned} E_i(q) =& \omega_{\text{touch}} T_i(q) + \omega_{\text{path}} P_i(q) \\ &+ \omega_{\text{schema}} A_i(q) + \omega_{\text{density}} \Delta_i(q), \end{aligned} \tag{2}$$

where $T_i(q)$ denotes candidate evidence contact, $P_i(q)$ denotes mechanism-oriented path support, $A_i(q)$ denotes type or schema compatibility, and $\Delta_i(q)$ denotes local evidence density. A contact-based instantiation uses $T_i(q) = \mathbb{I}[\deg_{S_q^{(0)}}(c_i) > 0]$, where $\deg_{S_q^{(0)}}(c_i)$ counts source-subgraph triples incident to the candidate. In this setting, $P_i(q)$ summarizes shared-gene overlap, gene–gene bridge evidence, and contacts

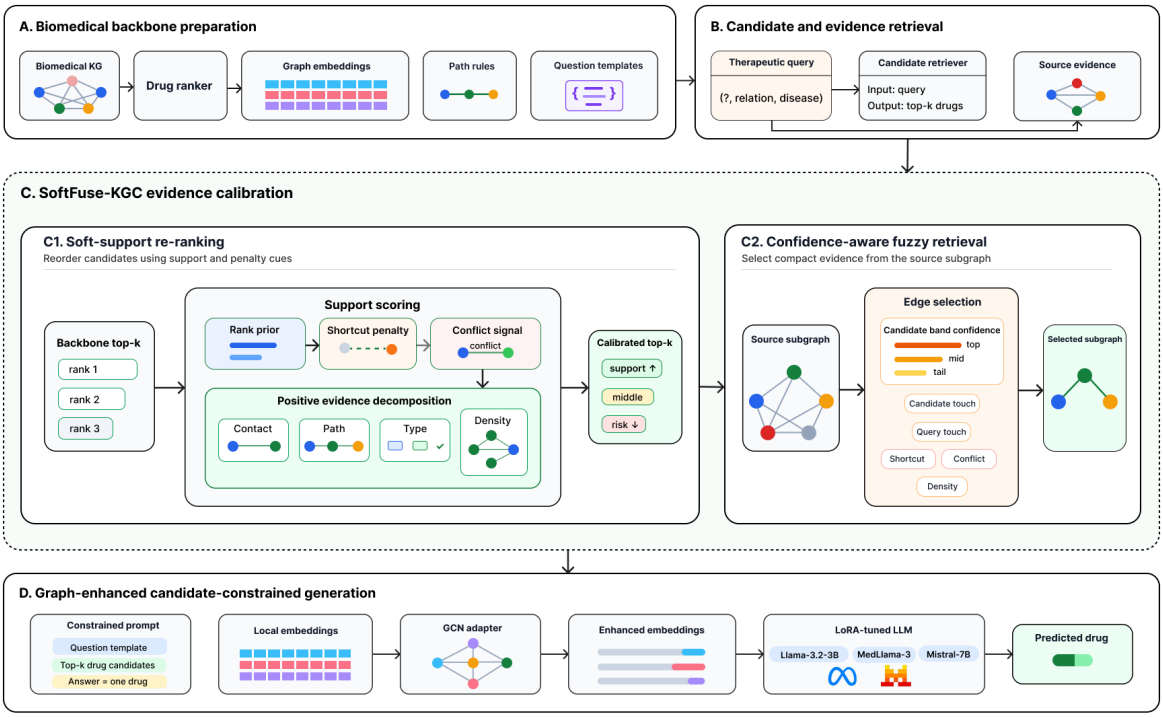


Fig. 1 SoftFuse-KGC framework for evidence-calibrated biomedical graph-augmented LLM completion. A. Biomedical backbone preparation builds the BKG, graph-compatible drug ranker, graph embeddings, path rules, and question templates. **B. Candidate and evidence retrieval** converts a therapeutic query into a top-20 candidate list and source evidence subgraph. **C. SoftFuse-KGC evidence calibration** includes C1 soft support reranking, which combines a rank prior, decomposed positive support, shortcut penalties, and contradiction signals, and C2 confidence-aware fuzzy retrieval, which selects compact evidence using candidate bands, candidate and query incidence, shortcut, conflict, and density cues. **D. Graph-enhanced candidate-constrained generation** passes the calibrated candidates and selected graph through the graph adapter and LoRA-tuned LLM to predict one drug

with candidate- and query-side mechanism neighborhoods:

$$\begin{aligned} P_i(q) =& \nu_s M_i^s(q) + \nu_b M_i^b(q) \\ &+ \nu_c T_i^c(q) + \nu_q T_i^q(q), \\ M_i^s(q) =& \min\left(1, \frac{|G(c_i) \cap G(q)|}{\eta_s}\right), \\ M_i^b(q) =& \min\left(1, \frac{B_i(q)}{\eta_b}\right), \end{aligned} \tag{3}$$

where $G(c_i)$ is the set of target or associated genes for the compound, $G(q)$ is the disease-gene set, and $B_i(q)$ counts gene–gene bridge edges between these sets. The terms $T_i^c(q)$ and $T_i^q(q)$ indicate whether the source subgraph contains candidate-side and query-side mechanism neighborhoods, respectively. Schema support is defined as follows:

$$A_i(q) = \mathbb{I}[\tau(c_i) \in \mathcal{T}_{\text{drug}}]\, \mathbb{I}[\tau(t_q) \in \mathcal{T}_{\text{disease}}], \tag{4}$$

when entity-type annotations are available. The density term is a bounded local count, for example,

$$\Delta_i(q) = \min\left(1, \frac{\log(1 + \deg_{S_q^{(0)}}(c_i) + \deg_{S_q^{(0)}}(t_q))}{\eta_\Delta}\right). \tag{5}$$

Together, these components make $E_i(q)$ a positive evidence score rather than a detector of a single direct link.

The two penalty terms represent different failure modes. The first, $D_i(q)$, captures a direct shortcut or candidate-level popularity:

$$D_i(q) = \max\left(\mathbb{I}[c_i \leftrightarrow t_q \text{ in } S_q^{(0)}], \min\left(1, \frac{\deg_{\text{direct}}(c_i)}{\eta_D}\right)\right), \tag{6}$$

The first component detects a direct candidate–query edge in the source graph. The second penalizes candidates that act as generic hubs across direct disease relations. The contradiction term $X_i(q)$ marks contraindication, conflict, failed, or suspended-indication evidence:

$$X_i(q) = \mathbb{I}[c_i \leftrightarrow t_q \text{ through contradiction-like evidence}]. \tag{7}$$

We first compute an evidence-only score and then, when selected, combine its within-query normalized value with the backbone rank prior:

$$\begin{aligned} \widetilde{S}_{\theta_d}(c_i, q) =& \alpha_{E,d} E_i(q) - \alpha_{D,d} D_i(q) - \alpha_{X,d} X_i(q), \\ S_{\theta_d}(c_i, q) =& \mu_d R_i(q) + (1 - \mu_d) N_q\Big(\widetilde{S}_{\theta_d}(c_i, q)\Big). \end{aligned} \tag{8}$$

Here, $R_i(q) = (K - r_i^{(0)} + 1)/K$ is the normalized backbone rank prior. The function $N_q(\cdot)$ is the within-query normalization when a rank-prior mixture is selected and the identity map otherwise. The index d denotes the dataset, and θ_d collects its calibration parameters.

Candidates are sorted by decreasing soft support score, with the original backbone rank used as a stable tiebreaker:

$$c_i \prec c_j \iff S_i > S_j \quad \text{or} \quad (S_i = S_j \text{ and } r_i^{(0)} < r_j^{(0)}), \tag{9}$$

where $S_i = S_{\theta_d}(c_i, q)$ and $r_i^{(0)} = \text{rank}_0(c_i)$. We denote the resulting candidate list by C_q^{soft}.

Online Resource 1 provides the feature-selection procedure, weight-assignment method, saturation constants, normalization choices, and validation-frozen coefficients for equations (2)–(8). Supplementary Section S1 describes the soft-support feature audit and coefficient assignment, Table S1 reports the dataset-specific instantiations, and Supplementary Section S6 summarizes the PrimeKG formula variants used as representative ablations.

Fig. 2 presents a validation-level view of this behavior on PrimeKG. Panel A compares candidate-ranking metrics for the raw backbone, ontology filtering, a contrastive support variant, and the selected soft support setting. Ontology filtering serves as the hard-filter control: candidates are retained only when they satisfy binary type, schema, or ontology-support conditions, while candidates without such support are discarded or demoted. This control is diagnostically useful but can be brittle in an incomplete BKG because missing ontology evidence is not equivalent to negative biomedical evidence. By contrast, the selected soft support setting retains all top-20 candidates and changes only their order. Panel B shows that this setting improves validation ranking while lowering the top-five direct-link rate. The comparison therefore supports the interpretation that the gain reflects graded evidence calibration rather than the promotion of shortcut-connected candidates.

3.4 Confidence-aware fuzzy retrieval

The fuzzy retrieval module changes the evidence graph supplied to the graph adapter while preserving a validation-selected candidate order. Let C_q^{in} denote this fixed input list. In the main pipeline, it is usually the soft-support list; the protocol can instead use a backbone-compatible list when validation analysis indicates that soft support should remain diagnostic rather than define the main row. Let $S_q^{(0)}$ denote the source evidence subgraph for query q, and let $e = (u, r, v)$ be an evidence triple. Fuzzy retrieval assigns a confidence-aware score to every edge and selects a compact subgraph S_q^{fuzzy} under a dataset-specific graph budget.

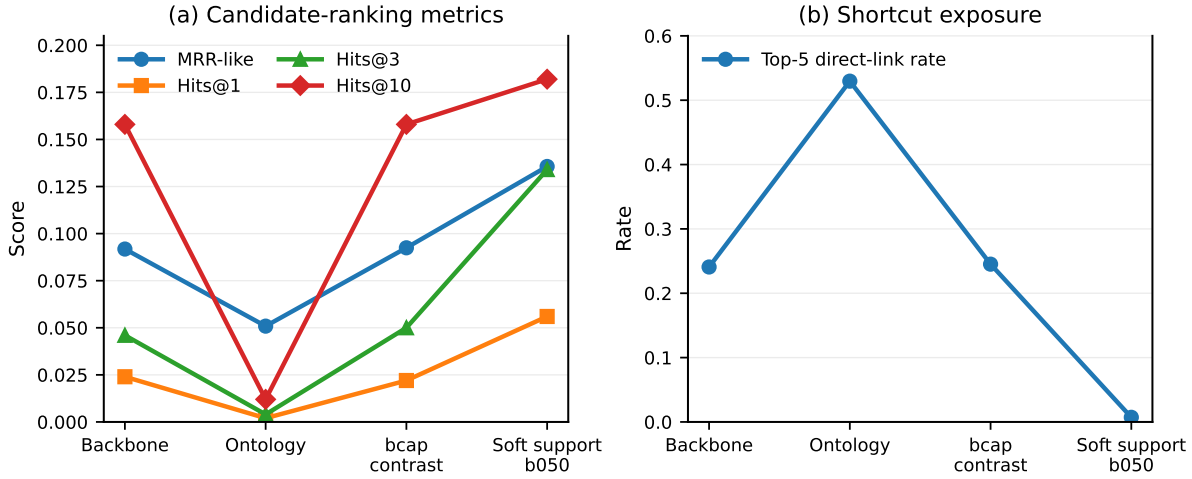


Fig. 2 Soft support calibration favors graded reranking over a hard filter. Panel (a) compares the raw DrKGC-style source, hard ontology filtering, a contrastive support variant, and the selected soft support b050 setting on PrimeKG validation candidate metrics. Panel (b) reports the top-five direct-link rate. The selected setting preserves Gold@20, improves rank quality, and reduces shortcut exposure, illustrating why the final score combines positive support $E_i(q)$ with shortcut and contradiction penalties instead of applying a binary keep-or-discard rule

A key component is candidate rank-band confidence. Under the fixed top-20 protocol, the selected list is partitioned into ordered bands for the highest-ranked, middle, and recoverable tail candidates. When an edge touches one or more candidates, its band score is the largest weight assigned to any touched candidate:

$$B(e,q) = \max_{c_i \in C_q^{\text{in}}: e \sim c_i} w_b(r_i), \tag{10}$$

and we set $B(e,q) = 0$ when e touches no candidate. Here, $e \sim c_i$ means that edge e is incident to candidate c_i or is annotated as part of a candidate-anchored evidence path; r_i is the selected rank of c_i.

The confidence-aware edge score combines rank-band confidence $B(e,q)$, candidate incidence $T_e^c(q)$, query incidence $T_e^q(q)$, a top-candidate bonus $H_e(q)$, a direct-shortcut penalty $D_e(q)$, a contradiction or failed-evidence penalty $X_e(q)$, local density $\rho_e(q)$, and relation-family priority $F_r(e)$. Dataset-specific instantiations give zero coefficients to unavailable or inactive terms.

The full edge score is computed as follows:

$$\begin{aligned} S_{\text{edge}}(e,q) =& \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\ &+ \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\ &- \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned} \tag{11}$$

where

$$\begin{aligned} T_e^c(q) &= \mathbb{I}[u \in C_q^{\text{in}} \text{ or } v \in C_q^{\text{in}}], \\ T_e^q(q) &= \mathbb{I}[u = t_q \text{ or } v = t_q], \\ H_e(q) &= \mathbb{I}[\exists c_i \in C_q^{\text{in}} : \ e \sim c_i \text{ and } r_i \le K_{\text{top}}], \end{aligned} \tag{12}$$

For edge $e = (u, r, v)$, K_{top} denotes the upper boundary of the top-candidate band. The term $F_r(e)$ encodes relation-family priority, $D_e(q)$ captures direct-shortcut evidence, and $X_e(q)$ captures contraindication, conflict, failed, or suspended-indication evidence. The density term is computed as follows:

$$\rho_e(q) = \frac{\log(1 + \deg_{S_q^{(0)}}(u)) + \log(1 + \deg_{S_q^{(0)}}(v))}{\eta_\rho}, \tag{13}$$

and is clipped to a bounded range in the implementation.

The selected graph is represented as follows:

$$S_q^{\text{fuzzy}} = \text{TopEdges}\left(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\right), \tag{14}$$

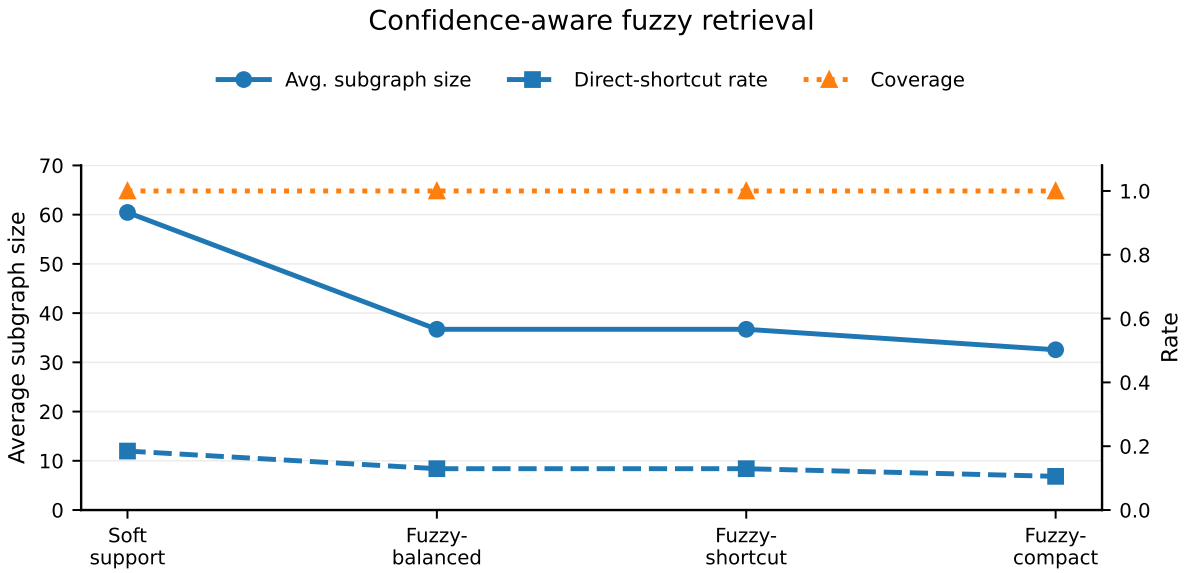


Fig. 3 Confidence-aware fuzzy retrieval selects the compact evidence package used after soft support. The diagnostic plot compares the soft-support source graph with fuzzy retrieval variants in terms of average subgraph size, direct-shortcut exposure, and candidate coverage. Because fuzzy retrieval preserves the soft-support candidate order, the selected variant is chosen for graph efficiency and evidence cleanliness while maintaining Gold@20, MRR@20, and candidate coverage

where $\tau_{d,q}$ is the dataset- and query-specific graph budget.

Edge selection follows the implementation order of a coverage-then-fill procedure. First, a candidate-coverage pass retains the highest-scoring available edge for each candidate that would otherwise lose all candidate-touching evidence, subject to the graph budget. When the prepared graph contains informative query-touching or mechanistic relation families, optional query-coverage and relation-family reservation passes are applied next. The remaining slots are filled in descending order of $S_{\text{edge}}(e, q)$ until the target size is reached. Fuzzy retrieval is therefore designed to produce a smaller, cleaner graph for the adapter and for human inspection, rather than to improve candidate MRR directly.

The fuzzy module preserves the order of its selected candidate list:

$$C_q^{\text{fuzzy}} = C_q^{\text{in}}. \tag{15}$$

Accordingly, candidate-stage changes are attributed to the input ranking, whereas graph-size reduction and evidence selection are attributed to fuzzy retrieval.

Online Resource 1 provides the feature-selection procedure, edge-weight assignment, rank-band definitions, relation-family priorities, top-candidate bonus settings, density normalizers, graph budgets, and coverage checks for equations (10)–(15). Supplementary Section S2 describes the feature audit and edge-weight assignment, Table S2 reports the frozen edge-scoring weights, and Supplementary Section S3 and Table S3 report graph budgets and coverage checks.

Fig. 3 summarizes the evidence-selection trade-off. Starting from the selected input package, the fuzzy variants show how much evidence can be removed while the candidate list remains fixed. The selected setting is therefore not chosen for a new candidate-ranking gain. Instead, it preserves Gold@20, candidate MRR, and candidate coverage while supplying a smaller graph to the adapter.

3.5 Graph-adapter-enhanced generation

For each query, the prompt represents the query entity and all candidate entities with special placeholders. The graph enhancer replaces their placeholder embeddings with graph-aware embeddings. Each entity begins with a frozen structural embedding from the lightweight ranker. When a subgraph is available

Table 1 Summary of the five biomedical evaluation datasets. Train, validation, and test values count target-relation triples. Candidates are the size of the restricted drug or compound universe. Graph edges, entities, and relations describe the prepared training/evidence graph after validation and test target edges were excluded.

Dataset	Target relation	Train	Valid	Test	Candidates	Graph edges	Entities	Relations
PrimeKG	indication	8,388	500	500	1,801	136,351	10,453	4
PharmKG	T proxy	28,960	500	500	1,342	386,768	7,247	28
Hetionet	CtD	555	100	100	1,552	2,249,997	47,031	24
DRKG	DrugBank treats	3,968	500	500	1,542	187,962	13,918	85
repoDB	approved indication	5,677	500	500	1,519	90,804	8,818	63

and sufficiently large, a relation-aware graph convolution updates node representations within the query-specific graph. The adapter then projects the concatenation of the global and locally updated embeddings to the hidden dimension of the LLM:

$$\mathbf{z}_e = \text{Adapter}\left([\mathbf{h}_e^{\text{global}}; \mathbf{h}_e^{\text{local}}]\right). \tag{16}$$

If an entity is absent from the retrieved subgraph, its local embedding is set to its global embedding. This fallback prevents missing graph evidence from producing an invalid representation and is consistent with the soft-evidence interpretation.

We fine-tune the LLM through LoRA [36] adapters while training the graph adapter and keeping the structural embeddings frozen. The prompt requires one candidate entity as the answer. Llama 3.2 3B is the primary language model for end-to-end evaluation.

4 Experimental Results

4.1 Datasets and task setting

We represent each BKG as a directed, multirelational graph of biomedical entities and typed relations. Across all five datasets, drug-focused therapeutic completion is formulated as head prediction. Given a query tail t_q, the system ranks a drug or compound head for the incomplete triple $(?, r_q, t_q)$, where r_q is a dataset-specific therapeutic or indication-style relation. Table 1 summarizes the target tasks, data splits, restricted candidate universes, and prepared training/evidence graphs. The graph statistics describe the evidence supplied to the graph-compatible pipeline after target triples assigned to validation or test were excluded; they are not the sizes of the unprocessed source KGs.

PrimeKG [5] is the primary evaluation setting. We derived a drug-restricted indication benchmark and prepared an evidence graph with four relation families: indication, target, associated_with, and protein–protein interaction. This combination supports both direct therapeutic context and mechanism-oriented drug–gene–disease paths. PharmKG [6] provides a transfer setting with a larger relation vocabulary. Its relation T is treated as a therapeutic-association proxy rather than a confirmed clinical-indication label, so results on this dataset indicate transfer across graph semantics rather than direct replication of the PrimeKG task.

Hetionet [4] uses the compound-treats-disease relation CtD, with the candidate universe restricted to compounds. Its evidence graph is much larger than the other prepared graphs and has a distinct topology and relation vocabulary, providing a test of robustness to graph scale and structure. For DRKG [7], we use the DrugBank [13] relation `DRUGBANK::treats::Compound:Disease` as the target. The controlled evidence graph combines selected compound–gene target relations, disease–gene relations, gene–gene relations, and auxiliary compound–disease relations rather than using every raw DRKG edge.

In repoDB [8], approved drug–indication pairs define positive target triples, whereas failed pairs are retained as diagnostic evidence. repoDB disease identifiers are UMLS [9] CUIs that do not map directly to the reused DRKG disease namespace, so they remain local UMLS nodes. The resulting graph combines approved target edges, failed-like diagnostic edges, and DRKG-derived compound–gene and gene–gene

evidence. This identifier mismatch produces sparser query-specific mechanistic paths and makes repoDB a useful test of evidence calibration under limited cross-resource alignment.

Together, the five settings vary in target semantics, candidate-universe size, graph density, relation diversity, and availability of mechanism-oriented support. These differences allow the evaluation to distinguish improvements that transfer across datasets from effects that depend on a particular graph topology or evidence family.

4.2 Evaluation metrics

All experiments use the same fixed top-20 protocol. Candidate lists for validation and test queries are generated before scoring, and a gold entity absent from a list receives the sentinel rank 21. We use $K = 20$ because this is the candidate-list size accepted by the graph-LLM backbone. It is also small enough for candidate-constrained generation while supporting informative Hits@1/3/10 and Gold@20 analyses.

Here, Q denotes the set of evaluation queries; y_q is the gold entity for query q; C_q is its fixed candidate list; and $\mathrm{rank}_C(y_q)$ is the one-based position of the gold entity in that list. A present gold entity receives its top-20 position, whereas an absent gold receives rank 21. Reciprocal rank is caculated as follows:

$$\mathrm{RR@20}(q) = \begin{cases} 1/\mathrm{rank}_C(y_q), & \mathrm{rank}_C(y_q) \le 20, \\ 0, & \mathrm{rank}_C(y_q) = 21. \end{cases} \tag{17}$$

The candidate MRR is caculated as follows:

$$\mathrm{MRR@20} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR@20}(q). \tag{18}$$

Hits@K and Gold@20 are caculated as follows:

$$\mathrm{Hits@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}\left[\mathrm{rank}_C(y_q) \le K\right], \quad K \in \{1, 3, 10\}, \tag{19}$$

$$\mathrm{Gold@20} = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}[y_q \in C_q]. \tag{20}$$

Hits@1, Hits@3, and Hits@10 measure first-choice accuracy, early-rank utility, and broader candidate-list usefulness, respectively. We report Gold@20 separately because candidate coverage is a prerequisite for candidate-constrained generation.

For end-to-end evaluation, the LLM output is normalized and matched against the candidate names. Let $\hat{y}_q$ denote the normalized generated answer. An exact match with the gold answer gives E2E rank 1, whereas an absent gold entity gives E2E rank 21. When the generated answer is incorrect but occurs above the gold entity in the candidate list, the gold rank is unchanged. In all other cases, the gold rank is displaced by one position:

$$r_{\mathrm{e2e}}(q) = \begin{cases} 1, & \hat{y}_q = y_q, \\ 21, & y_q \notin C_q, \\ \mathrm{rank}_C(y_q), & \hat{y}_q \in C_q, \\ & \mathrm{rank}_C(\hat{y}_q) < \mathrm{rank}_C(y_q), \\ \min(21, \mathrm{rank}_C(y_q) + 1), & \text{otherwise.} \end{cases} \tag{21}$$

Table 2 Fixed top-20 candidate-ranking comparison across five biomedical datasets. All values are test metrics under the same absent-gold rank-21 rule. Metrics marked with (↑) are better when larger. Bold marks the best value for each dataset and metric. SoftFuse-KGC is the main graph-compatible row and should be interpreted as an evidence-calibrated DrKGC-style source, not as a universal replacement for the strongest structure-only candidate generator.

Model	Metric	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
TransE	Gold@20 (↑)	0.104	0.060	0.270	0.270	0.248
	MRR@20 (↑)	0.022	0.009	0.046	0.055	0.053
	Hits@10 (↑)	0.072	0.028	0.150	0.160	0.130
DistMult	Gold@20 (↑)	0.434	**0.102**	0.390	**0.432**	0.482
	MRR@20 (↑)	0.089	0.017	0.072	**0.100**	**0.104**
	Hits@10 (↑)	0.264	0.038	**0.290**	**0.276**	**0.320**
ComplEx	Gold@20 (↑)	**0.574**	0.066	**0.440**	0.406	**0.502**
	MRR@20 (↑)	0.125	0.009	0.090	0.086	0.084
	Hits@10 (↑)	**0.384**	0.036	0.250	0.260	0.312
RotatE	Gold@20 (↑)	0.134	0.024	0.200	0.182	0.202
	MRR@20 (↑)	0.033	0.002	0.043	0.024	0.022
	Hits@10 (↑)	0.074	0.004	0.120	0.098	0.100
R-GCN	Gold@20 (↑)	0.220	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	0.060	0.020	0.099	0.043	0.048
	Hits@10 (↑)	0.186	0.046	0.220	0.120	0.128
HRGAT	Gold@20 (↑)	0.212	0.094	0.410	0.204	0.222
	MRR@20 (↑)	0.033	0.019	**0.101**	0.042	0.052
	Hits@10 (↑)	0.128	0.044	**0.290**	0.134	0.146
DrKGC-style source	Gold@20 (↑)	0.240	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	0.065	0.020	0.099	0.043	0.048
	Hits@10 (↑)	0.192	0.046	0.220	0.120	0.128
SoftFuse-KGC	Gold@20 (↑)	0.240	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	**0.125**	**0.028**	0.095	0.047	0.048
	Hits@10 (↑)	0.222	**0.068**	0.280	0.142	0.128

The end-to-end reciprocal rank is described as below:

$$\mathrm{RR@20}_{\mathrm{e2e}}(q) = \begin{cases} 1/r_{\mathrm{e2e}}(q), & r_{\mathrm{e2e}}(q) \le 20, \\ 0, & r_{\mathrm{e2e}}(q) = 21. \end{cases} \tag{22}$$

This adjusted rank definition is conservative for candidate-constrained generation: it gives credit only when the generated answer matches the gold or when the candidate ordering still places the gold in an evaluated position.

4.3 Candidate stage comparison with structure baselines

The first analysis evaluates candidate retrieval before LLM generation. This stage determines whether candidate-constrained generation can recover the gold drug. A gold entity outside the fixed top-20 list cannot be selected, while a gold entity near the tail remains difficult for both the model and a human reviewer to identify. Table 2 therefore reports three complementary quantities. Gold@20 measures recoverability, MRR@20 measures gold position under the absent-gold rank-21 rule, and Hits@10 indicates whether the gold appears in the upper half of the list.

The table contains two types of rows. The standalone structure-only baselines provide reference candidate-retrieval performance for conventional embedding and graph models. The DrKGC-style source and SoftFuse-KGC rows are graph-compatible: in addition to ranking candidates, they must provide identifiers, embeddings, and query-specific source subgraphs for the graph adapter. Both graph-compatible rows use the same lightweight R-GCN-compatible source. Their difference therefore reflects evidence calibration rather than replacement of the upstream retriever.

PrimeKG provides the clearest candidate-stage result. SoftFuse-KGC increases MRR@20 from 0.065 to 0.125 while Gold@20 remains 0.240, showing that soft support reorders recoverable candidates rather than expanding the set. PharmKG follows the same direction in the transfer setting: MRR@20 rises from

Table 3 End-to-end module comparison for the graph-LLM pipeline. Gold@20, candidate MRR@20, and E2E MRR@20 are higher-is-better metrics (↑), whereas average graph size is lower-is-better (↓). E2E MRR@20 uses Llama 3.2 3B candidate-constrained generation with the adjusted-rank rule defined in Section 4.2. Bold marks the best row for each dataset and metric where the comparison is meaningful.

Row	Metric	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
DrKGC-style source	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	0.0646	0.0205	**0.0994**	0.0432	**0.0481**
	E2E MRR@20 (↑)	0.0476	0.0156	0.1045	0.0527	0.0487
	Avg. graph (↓)	59.93	100.00	80.00	100.00	100.00
Soft support	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	**0.1253**	**0.0282**	0.0954	**0.0466**	0.0481
	E2E MRR@20 (↑)	0.0747	0.0206	**0.1109**	0.0514	0.0509
	Avg. graph (↓)	59.93	100.00	80.00	100.00	100.00
Fuzzy retrieval	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	**0.1253**	**0.0282**	0.0954	**0.0466**	**0.0481**
	E2E MRR@20 (↑)	**0.0747**	**0.0210**	0.0991	**0.0549**	**0.0529**
	Avg. graph (↓)	**32.34**	**55.00**	**44.00**	**55.00**	**55.00**

0.020 to 0.028 and Hits@10 from 0.046 to 0.068, with Gold@20 unchanged at 0.092. DRKG shows a smaller positive MRR@20 change, from 0.043 to 0.047. Hetionet is mixed. Gold@20 remains 0.440 and Hits@10 increases from 0.220 to 0.280, but MRR@20 decreases from 0.099 to 0.095. Soft support therefore improves some upper-half placements while moving a small number of first-rank cases downward. repoDB is neutral at the candidate stage, consistent with its local UMLS disease nodes and sparse query-specific mechanistic support.

The standalone results provide context for these changes. ComplEx achieves the highest PrimeKG Gold@20; HRGAT and R-GCN are strong on Hetionet; and DistMult has the highest candidate MRR@20 on DRKG and repoDB. SoftFuse-KGC should therefore not be interpreted as a replacement for the strongest structure-only retriever. Its specific role is to calibrate the graph-compatible candidate package passed to the graph-LLM pipeline. The next analysis tests whether candidate calibration and evidence selection are reflected in end-to-end behavior.

4.4 End-to-end graph-LLM pipeline comparison

Table 3 compares the three graph-compatible pipeline rows after candidate-constrained generation: the raw DrKGC-style source, the soft-support row, and the confidence-aware fuzzy retrieval row. This system-level comparison reflects both inputs available to the downstream model: an ordered candidate list and a query-specific graph package delivered through the adapter. It also isolates the proposed modules. Soft support changes candidate order, whereas fuzzy retrieval preserves that order and changes the evidence graph.

PrimeKG again shows the clearest improvement. Soft support raises E2E MRR@20 from 0.0476 to 0.0747, consistent with the candidate-stage change. Fuzzy retrieval preserves this value while reducing the average graph from 59.93 to 32.34 triples. Thus, candidate calibration makes recoverable gold drugs more visible, after which fuzzy retrieval removes nearly half of the graph context without losing the gain. PharmKG provides a transfer result in the same direction. Although relation T is a therapeutic association proxy rather than a confirmed clinical indication, E2E MRR@20 increases from 0.0156 to 0.0210 while the graph shrinks from 100 to 55 triples.

The external datasets show the limits of this pattern. On Hetionet, soft support produces the best test E2E MRR@20, increasing it from 0.1045 to 0.1109 even though candidate MRR@20 is slightly below the raw source. A graph-augmented LLM may therefore benefit from a more mechanism-aware candidate context even when the candidate-only metric does not improve. Fuzzy retrieval, however, lowers Hetionet E2E MRR@20 to 0.0991 while reducing the graph from 80 to 44 triples. In this topology, graph compression trades off with end-to-end behavior. On DRKG and repoDB, DistMult remains the stronger standalone candidate generator, but fuzzy retrieval still improves the graph-compatible pipeline: E2E

Table 4 PrimeKG diagnostic cases of candidate promotion and evidence compression. Base and Soft denote gold-drug ranks before and after soft-support reranking; Removed is the number of triples discarded by fuzzy retrieval. These cases audit rank and evidence changes rather than provide treatment recommendations.

Bucket	Query / gold drug	Base	Soft	Removed	Interpretation
Improved	lymphosarcoma / Vincristine	16	2	20	Soft support promotes an evidence-connected antineoplastic candidate, and fuzzy retrieval preserves the improved order.
Improved	lymphoma / Vincristine	17	3	20	The gold drug is promoted from the tail under related lymphoma terminology.
Improved	rheumatoid arthritis / Methotrexate	19	8	23	The benchmark gold drug moves upward, showing why Gold@20 and MRR@20 should be reported together.
Sparse	seborrheic dermatitis / Cortisone acetate	1	1	26	Soft scoring avoids strict filtering when graph evidence is incomplete.
Cleaner	chronic otitis media / Norfloxacin	4	4	38	Candidate order is preserved while the graph package is substantially reduced.
Cleaner	cholera / Fusidic acid	1	1	33	Fuzzy retrieval removes redundant or shortcut-heavy evidence.
Cleaner	botulism / Ofloxacin	12	10	30	Candidate ordering is mostly preserved with a cleaner evidence graph.

MRR@20 rises from 0.0527 to 0.0549 on DRKG and from 0.0487 to 0.0529 on repoDB as graph size falls from 100 to 55 triples.

These findings support two related but distinct conclusions. Soft support is most useful when the prepared graph contains query-specific features that distinguish candidates, as on PrimeKG and PharmKG and, more modestly, DRKG. Its effect is weaker when support is sparse or graph topology differs, as in repoDB and parts of Hetionet. Fuzzy retrieval primarily improves evidence efficiency. By design, it preserves the candidate MRR of its input row, while reducing graph size by approximately 45% in every fuzzy setting. The graph adapter consequently receives a smaller and more inspectable context, and retained evidence is easier to audit. The repoDB result indicates relevance to a clinical benchmark rather than prospective clinical validation: the dataset includes approved pairs and failed-like evidence, but its local UMLS disease nodes limit query-specific mechanistic support.

4.5 Case studies

The case studies examine whether the numerical changes in Table 3 correspond to evidence changes that a biomedical reader can inspect. A correct graph-augmented LLM answer may arise in several ways: the candidate generator may already rank the gold drug highly, the LLM may select an already high-ranked candidate, or evidence calibration may promote a recoverable but low-ranked candidate. We therefore examine representative PrimeKG examples in terms of rank change, graph-size reduction, retained internal evidence, and external biomedical plausibility.

Table 4 groups the cases into three diagnostic categories. Improved cases test whether soft support promotes recoverable, low-ranked gold drugs to more visible positions. Sparse cases test whether graded scoring avoids brittle hard filtering when graph support is incomplete. Cleaner cases assess fuzzy graph reduction when candidate ranks change little. This taxonomy follows the division of labor within SoftFuse-KGC: soft support targets candidate order, whereas fuzzy retrieval targets evidence efficiency. Fig. 4 visualizes the two principal examples, and Table 5 reports external evidence identifiers for model-selected candidates.

The first principal case is lymphosarcoma / vincristine. Vincristine occurs in the raw DrKGC-style top-20 list but only at rank 16. Soft support promotes it to rank 2, after which fuzzy retrieval preserves the improved order while reducing the evidence subgraph from 45 to 25 triples. This example is informative because "lymphosarcoma" is a historical lymphoma-related term. Fig. 4(a) shows that fuzzy retrieval retains the promoted vincristine context and additional lymphoma-relevant local evidence while removing redundant triples.

The second principal case is rheumatoid arthritis / methotrexate. Clinical guidance, labeling, and clinical studies identify methotrexate as an established rheumatoid arthritis therapy [56, 61, 62]. The

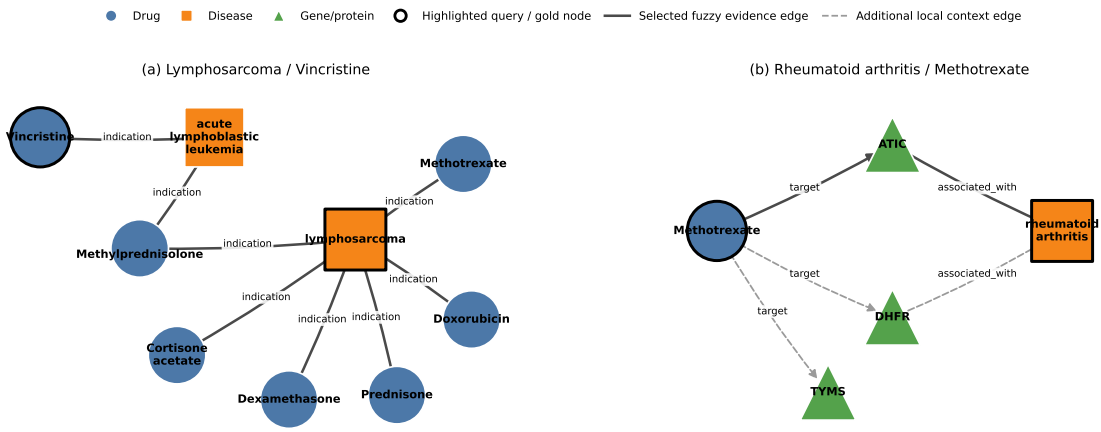


Fig. 4 Case-study evidence graphs for lymphosarcoma / Vincristine and rheumatoid arthritis / Methotrexate. Circles denote drugs, squares denote diseases, and triangles denote genes or proteins. Thick solid edges show evidence selected by fuzzy retrieval, whereas dashed edges show additional local context. Black outlines identify the query disease and gold drug, making rank changes and retained evidence auditable

Table 5 Internal graph evidence and external plausibility evidence for case-study candidates. Candidate drugs come from the corresponding SoftFuse-KGC ranked lists, and internal evidence is extracted from the fuzzy-retrieved graph. The external evidence column provides a literature citation and PubMed identifier for candidate–disease plausibility or related clinical context. These references support plausibility checking and auditability, not prospective clinical validation, graph-edge provenance, or treatment recommendations.

Case disease	Candidate drug	Model rank / band	Internal SoftFuse-KGC evidence	External plausibility evidence
Lymphosarcoma	Prednisolone	Rank 1 / top	Prednisolone – indication – mycosis fungoides and variants	[51]; PMID:8422281
Lymphosarcoma	Vincristine	Rank 2 / top	Vincristine – indication – acute lymphoblastic leukemia	[52]; PMID:5066691
Lymphosarcoma	Dexamethasone	Rank 5 / top	Dexamethasone – indication – lymphosarcoma; reverse direct edge retained	[53]; PMID:8120556
Lymphosarcoma	Doxorubicin	Rank 12 / tail	Doxorubicin – indication – lymphosarcoma	[54]; PMID:7680764
Lymphosarcoma	Bleomycin	Rank 15 / tail	Bleomycin – indication – lymphosarcoma	[55]; PMID:1384144
Rheumatoid arthritis	Methotrexate	Rank 8 / mid	Methotrexate – target – ATIC; ATIC – associated_with – rheumatoid arthritis	[56]; PMID:34101387
Rheumatoid arthritis	Dexamethasone	Rank 11 / tail	Dexamethasone – indication – rheumatoid arthritis	[57]; PMID:4158821
Rheumatoid arthritis	Hydrocortisone	Rank 13 / tail	Hydrocortisone – indication – rheumatoid arthritis	[58]; PMID:8173851
Rheumatoid arthritis	Prednisolone	Rank 16 / tail	Prednisolone – indication – rheumatoid arthritis	[59]; PMID:35641125
Rheumatoid arthritis	Prednisone	Rank 18 / tail	Prednisone – indication – rheumatoid arthritis	[60]; PMID:18207016

raw candidate generator retrieves methotrexate at rank 19, a weak position for candidate-constrained generation and manual review. Soft support raises it to rank 8, and fuzzy retrieval preserves that rank while reducing the graph from 52 to 29 triples. Fig. 4(b) retains the mechanistic Methotrexate–ATIC–rheumatoid arthritis path together with additional target and association context. The example illustrates why Gold@20 and MRR@20 should be considered together: the raw list contains the gold drug, but its position limits practical utility.

The external evidence in Table 5 provides a plausibility check for model-selected candidates rather than provenance for every internal graph edge. In the lymphosarcoma case, the table includes the promoted gold drug Vincristine and other lymphoma-context candidates ranked by the model. In the rheumatoid arthritis case, methotrexate is the gold candidate, whereas the listed steroids represent adjunctive anti-inflammatory context. The candidates are not claimed to be therapeutically equivalent.

The remaining cases illustrate complementary behavior. Cortisone acetate remains at rank 1 for seborrheic dermatitis and shows why binary filtering can be risky when graph support is incomplete. The chronic otitis media / norfloxacin, cholera / fusidic acid, and botulism / ofloxacin examples show that fuzzy retrieval can remove redundant or shortcut-heavy triples while preserving candidate order. Because graph proximity does not itself establish therapeutic appropriateness, all cases require cautious biomedical interpretation. Their purpose is to show that SoftFuse-KGC makes rank and evidence changes auditable, not to provide clinical validation.

5 Discussion

The results support soft, confidence-aware evidence modeling for biomedical graph-augmented LLM completion. More specifically, they show that candidate ordering and graph-evidence selection are separate problems. Soft support can improve ranking when the available evidence differentiates candidates, whereas fuzzy retrieval can reduce graph size without changing the candidate list. This distinction matters in biomedical informatics because ranking performance, evidence efficiency, and interpretability all affect the usefulness of a graph-LLM pipeline.

Evaluation across five datasets shows that the benefit of soft support depends on the evidence available for each query. PrimeKG and PharmKG show clear candidate-stage improvements, and DRKG shows a smaller gain. Hetionet and repoDB are neutral or mixed. This variation clarifies the distinct roles of the two modules. Soft support relies on query-specific signals that distinguish one candidate from another; when disease–gene or mechanism-oriented evidence is sparse, its scores may become too generic to support reliable reranking. Fuzzy retrieval has a different objective. Even when candidate ranking does not improve, it can preserve the selected order while reducing the evidence package. The Hetionet result also shows that excessive compression may remove context useful to the generator, so graph budgets require dataset-specific validation.

The structure-only comparisons further define the scope of the contribution. R-GCN serves as the lightweight source because it matches the DrKGC graph-adapter interface and provides relation-aware local embeddings for the subgraph encoder. It is not assumed to be the strongest standalone candidate generator. The calibrated graph-compatible row is strongest or among the strongest in candidate MRR on PrimeKG and PharmKG, whereas HRGAT or DistMult performs better on Hetionet, DRKG, and repoDB. These results indicate the limitation imposed by the selected source but also illustrate the modularity of SoftFuse-KGC. A future implementation could replace R-GCN with a stronger structural or hybrid retriever that exposes compatible entity embeddings, candidate lists, and query-specific evidence graphs.

repoDB is a clinically relevant external benchmark rather than clinical validation. Its approved drug–indication pairs and failed-like diagnostic evidence are useful for evaluation, but its local UMLS disease nodes do not map directly to the reused DRKG disease namespace. Query-specific mechanistic paths are therefore sparse, which limits support reranking. Nevertheless, fuzzy retrieval preserves candidate metrics, reduces the average graph from 100 to 55 triples, and yields the best R-GCN-derived E2E MRR. This result supports the evidence-efficiency claim under a difficult mapping setting while also demonstrating the limits of soft support.

The case studies make these behaviors more interpretable, but they require careful biomedical framing. The lymphosarcoma / vincristine example combines a large rank improvement with retained lymphoma-related evidence. The rheumatoid arthritis / methotrexate example combines a rank improvement with guideline, labeling, clinical-study, and graph-path context. These examples demonstrate candidate reranking and evidence packaging rather than clinical validity. SoftFuse-KGC should therefore be used for prioritization, hypothesis generation, and evidence review. It does not replace clinical judgment, pharmacological assessment, or experimental validation.

The current study also has several limitations. First, the top-20 candidate bottleneck is fundamental: candidate-constrained generation cannot recover an absent gold drug. Second, stronger structural

or hybrid retrievers may improve coverage if they provide compatible embeddings, candidate lists, and evidence graphs. Third, support calibration is validation-selected and manually specified; learned or expert-curated weights may improve robustness. Fourth, fuzzy retrieval uses score-based edge selection rather than a learned retriever. Fifth, relation semantics differ across the external datasets: PharmKG relation T is a therapeutic association proxy, the DRKG DrugBank relation is not identical to the PrimeKG indication relation, and repoDB approved pairs form a clinical benchmark. Finally, the case studies are qualitative and require domain-expert review.

6 Conclusion

In this paper, we introduced SoftFuse-KGC, a soft support and confidence-aware fuzzy retrieval framework for biomedical graph-augmented LLM completion. The method extends a DrKGC-style backbone for drug-focused head prediction with two evidence-aware modules: a soft support reranker for candidate ordering and a fuzzy retriever for compact graph selection. Across five datasets, SoftFuse-KGC improved candidate ordering in several settings, preserved the selected ranking while reducing graph size in every fuzzy retrieval setting, and revealed dataset-dependent limits associated with sparse support. Calibrated evidence modeling is therefore useful when candidate coverage, graph noise, and interpretability must be considered together in therapeutic hypothesis generation.

Future work will pursue several directions. First, we will integrate stronger structural and hybrid candidate retrievers to improve candidate coverage while retaining compatibility with the graph-adapter interface. Second, we will investigate learned, uncertainty-aware support calibration and query-adaptive graph budgets as alternatives to manually specified coefficients and fixed retrieval sizes. Third, we will strengthen cross-dataset entity and ontology alignment, particularly between repoDB UMLS concepts and mechanistic graph namespaces, and extend the evaluation to additional therapeutic relations. Fourth, learned evidence retrievers will be studied to balance graph compression against end-to-end performance across different graph topologies. Finally, temporal evaluation, broader language-model comparisons, and domain-expert assessment will be used to examine whether the retained evidence remains robust, mechanistically informative, and clinically plausible.

Supplementary Information. Additional details are provided in the Supplementary Material. The opening *The Web Application* section describes the public demo, screenshots, and tab-level usage notes. We provide a public web application for interactive inspection of the SoftFuse-KGC pipeline at https://inosuke710-softfuse-kgc-demo.hf.space/. It uses precomputed artifacts and allows users to select a dataset, split, disease or query case, and top-k value before viewing the Overview, Soft Support, and Fuzzy Retrieval tabs. Across tabs, the interface links a selected case to summary metrics, candidate-level support features, rank changes, and retained evidence edges. Users can therefore trace a reranking or graph-compression result to the final evidence package. Sections S1–S3 report feature selection, weight assignment, frozen soft-support coefficients, fuzzy edge-scoring weights, and graph-budget checks. Sections S4–S6 provide the inference algorithm, graph-LLM settings, prompt and retrieval-rule templates, and PrimeKG formula variants. Sections S7–S9 report LLM comparisons, decoding checks, ablation, sensitivity, robustness, and the fuzzy encoder probe. Sections S10–S11 provide structure-only baseline training settings, dataset preprocessing, and leakage checks. Tables S1–S15 contain the corresponding configuration and diagnostic summaries.

Statements and Declarations

Competing interests
Authors have no conflict of interest relevant to this article.

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Ethics statement
This study used publicly available biomedical knowledge graph resources and involved no human participants, human tissue, animal experiments, or identifiable patient-level data.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process
During the preparation of this work, the authors used ChatGPT (OpenAI) for assistance with grammar checking, language polishing, and minor improvements to sentence structure. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

Author contributions
Le Van Thanh: Conceptualization, Methodology, Investigation, Resources, Writing-original draft, Validation, Writing-review and editing; Dang Hoang Duong: Conceptualization, Methodology, Investigation, Writing-original draft, Visualization, Software, Hai Van Pham: Writing-review and editing, Validation, Supervision, Project administration.. All authors read and approved the final manuscript.

Data Availability Statement
These datasets are publicly available and can be accessed from their original public sources. The links to the datasets are as follows:

PrimeKG: https://doi.org/10.7910/DVN/IXA7BM (raw CSV: https://dataverse.harvard.edu/api/access/datafile/6180620);

PharmKG: https://github.com/biomed-AI/PharmKG and https://zenodo.org/records/4077338;

Hetionet v1.0: https://zenodo.org/records/268568;

DRKG at https://github.com/gnn4dr/DRKG (archive: https://dgl-data.s3-us-west-2.amazonaws.com/dataset/DRKG/drkg.tar.gz);

repoDB: https://github.com/adam-sam-brown/repoDB.

Code availability
The source code, figure-generation scripts, processed artifacts, and reproduction instructions are available at https://github.com/dhduong710/SoftFuse-KGC.

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